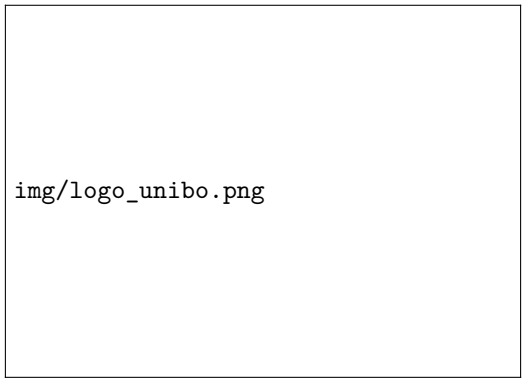


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Theoretical Physics

Advanced Quantum Field Theory

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Introduction

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1 | Gauge Theories

Gauge theories are building blocks of the Standard Model of particle physics. They arise from the requirement that massless spin-one particles, which mediate some of the fundamental forces of nature, should carry only two independent polarizations[1] even when described in terms of equations of motion that are manifestly Lorentz invariant: the photon is conveniently described by the vector field $A_\mu(x)$, which has four independent components as any four-vector, though the expected physical degrees of freedom are only two. The other two degrees of freedom are redundant because of the gauge symmetry: one can eliminate them by performing gauge transformations to reach a unitary gauge, such as the Coulomb gauge used in electrodynamics. However, in such a gauge, the Lorentz symmetry is not manifest, and it becomes difficult to exploit its consequences in a useful manner. It is sometimes more convenient to keep some redundancy associated with the gauge symmetry and maintain the Lorentz invariance as manifest as possible. Then, the principle of gauge invariance can be used to fix in a simple way all possible interactions mediated by massless particles of spin one in a way that is consistent with Lorentz invariance. Here, we introduce the principle of gauge invariance and use it to streamline the construction of the QED and QCD lagrangians, the main examples of abelian and non-abelian gauge theories, respectively.

[1] As implied by Wigner's classification of the irreducible unitary representations of the Poincaré group.

1.1 | Abelian gauge theories and QED

1.1.1 | The Free Dirac Field and Global Symmetry

Let us first review how, starting from the theory of free electrons described by the free Dirac equation, one elegantly deduces the complete Quantum Electrodynamics (QED) lagrangian. This derivation relies heavily on the principle of *gauge invariance* and the fundamental requirement of *Lorentz invariance*. We begin by considering the lagrangian of a free Dirac field of mass m:

$$\mathcal{L}_{Dirac} = -\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi - m\bar{\psi}\psi \tag{1.1.1}$$

which is explicitly constructed to be Lorentz invariant, and thus perfectly consistent with special relativity.

Throughout this text, we employ a mostly plus metric convention, denoting the Minkowski metric by $\eta_{\mu\nu}$. The gamma matrices are taken to satisfy the fundamental Clifford algebra relation:

$$\{\gamma^{\mu}, \gamma^{\nu}\} = 2\eta^{\mu\nu}\,, \tag{1.1.2}$$

while the conjugate Dirac spinor $\bar{\psi}$ is formally defined by:

$$\bar{\psi} = \psi^{\dagger}\beta\,, \qquad \beta = i\gamma^{0}\,, \qquad (\gamma^{\mu})^{\dagger} = -\beta\gamma^{\mu}\beta\,. \tag{1.1.3}$$

One may straightforwardly check that the Dirac lagrangian is formally real up to total derivatives and, of course, Lorentz invariant. Crucially, it is also invariant under global internal symmetry transformations belonging to the Abelian group $U(1)$:

$$\begin{aligned} \psi(x) \quad &\rightarrow \quad \psi'(x) = e^{i\alpha}\psi(x)\,, \qquad e^{i\alpha} \in U(1) \\ \bar{\psi}(x) \quad &\rightarrow \quad \bar{\psi}'(x) = e^{-i\alpha}\bar{\psi}(x)\,. \end{aligned} \tag{1.1.4}$$

This is classified as a global symmetry because the parameter α is a constant, entirely independent of spacetime coordinates.

1.1.2 | Local Gauge Invariance and the Covariant Derivative

Let us now explore how to extend this symmetry to a local one by introducing arbitrary spacetime-dependent functions $\alpha(x)$:

$$\psi(x) \quad \rightarrow \quad \psi'(x) = e^{i\alpha(x)}\psi(x) \tag{1.1.5}$$

$$\bar{\psi}(x) \quad \rightarrow \quad \bar{\psi}'(x) = e^{-i\alpha(x)}\bar{\psi}(x)\,. \tag{1.1.6}$$

The mass term in the lagrangian is trivially invariant since the opposite phases cancel out perfectly:

$$m\bar{\psi}\psi \quad \rightarrow \quad m\bar{\psi}'\psi' = m\bar{\psi}e^{-i\alpha(x)}e^{i\alpha(x)}\psi = m\bar{\psi}\psi\,, \tag{1.1.7}$$

however, the term containing the partial derivative is not invariant:

$$\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi \quad \rightarrow \quad \bar{\psi}'\gamma^{\mu}\partial_{\mu}\psi' = \bar{\psi}e^{-i\alpha(x)}\gamma^{\mu}\partial_{\mu}(e^{i\alpha(x)}\psi) = \bar{\psi}\gamma^{\mu}\partial_{\mu}\psi + i\bar{\psi}\gamma^{\mu}\psi\partial_{\mu}\alpha(x)\,. \tag{1.1.8}$$

The local transformation produces an extra term that vanishes only for a constant $\alpha(x)$. Consequently, the free lagrangian is not invariant, and it must be modified to achieve gauge invariance

(invariance for arbitrary functions $\alpha(x)$). Incidentally, one may note that the term multiplying the derivative of $\alpha(x)$ is directly proportional to the Noether current $J^\mu = i\bar{\psi}\gamma^\mu\psi$ associated with the global symmetry discussed in (1.1.4).

To rigorously construct gauge invariant actions, it is extremely useful to introduce a formalism based on the definition of *tensors of the gauge group* and their corresponding **covariant derivatives**. The latter are constructed specifically to produce tensors out of tensors, preserving the transformation properties.

To start with, we define $\psi(x)$ as a tensor under the gauge group $U(1) = \{e^{i\alpha(x)}\}$ if it transforms exactly as in (1.1.5). It follows that $\partial_\mu\psi(x)$ is not a tensor, as its transformation rule is more complicated (as seen above). The covariant derivative acting on ψ is obtained by introducing a new gauge field A_μ (often referred to in differential geometry as a "connection"). It is defined by:

$$D_\mu = \partial_\mu - iA_\mu(x)\,. \tag{1.1.9}$$

To ensure the derivative behaves properly, one requires $A_\mu(x)$ to transform in a specific way so that the "tensorial" transformation rule remains valid for the derivative:

$$D_\mu\psi(x) \quad \to \quad D'_\mu\psi'(x) = e^{i\alpha(x)}D_\mu\psi(x) \tag{1.1.10}$$

where $D'_\mu = \partial_\mu - iA'_\mu(x)$.

With a short algebraic calculation we can find the transformation rule for the gauge field:

$$\begin{aligned} D'_\mu\psi'(x) &= (\partial_\mu - iA'_\mu(x))e^{i\alpha(x)}\psi(x) \\ &= e^{i\alpha(x)}\partial_\mu\psi(x) + i(\partial_\mu\alpha(x))e^{i\alpha(x)}\psi(x) - iA'_\mu(x)e^{i\alpha(x)}\psi(x), \\ e^{i\alpha(x)}D_\mu\psi(x) &= e^{i\alpha(x)}(\partial_\mu - iA_\mu(x))\psi(x) \\ &= e^{i\alpha(x)}\partial_\mu\psi(x) - iA_\mu(x)e^{i\alpha(x)}\psi(x), \end{aligned}$$

which after equating the two expressions and simplifying yields the following requirement for the gauge field:

$$A_\mu(x) \quad \to \quad A'_\mu(x) = A_\mu(x) + \partial_\mu\alpha(x)\,. \tag{1.1.11}$$

Armed with covariant derivatives, it becomes simple to obtain a fully gauge invariant lagrangian from our starting point in (1.1.1):

$$\boxed{\mathcal{L} = -\bar{\psi}\gamma^\mu D_\mu\psi - m\bar{\psi}\psi}\,. \tag{1.1.12}$$

Charge of the Representation and Generalization of the Covariant Derivative

A "tensor" for the gauge group $U(1)$ is more generally defined as a field $\psi_q(x)$ that transforms as

$$\psi_q(x) \to \psi'_q(x) = e^{iq\alpha(x)}\psi_q(x), \tag{1.1.13}$$

where the integer $q \in \mathbb{Z}$ is called the "charge" of $\psi_q(x)$. Thus, $\psi_q(x)$ is considered a tensor of charge q: in more formal mathematical terms, q identifies a specific irreducible representation of the group $U(1)$.

The general definition of the covariant derivative is then naturally extended to:

$$D_\mu = \partial_\mu - iA_\mu(x)Q \tag{1.1.14}$$

where Q is an operator that measures the charge of the tensor on which it acts, meaning it functions as the generator of the $U(1)$ group in the representation of the target tensor. We can now easily recognize that the transformation of the conjugate spinor in (1.1.6) corresponds precisely to that of a tensor of charge -1, meaning the covariant derivative of $\bar{\psi}$ reads:

$$D_\mu \bar{\psi}(x) = (\partial_\mu + iA_\mu(x))\bar{\psi}(x)\,, \tag{1.1.15}$$

which is completely consistent with taking the Dirac conjugate of $D_\mu \psi$. The covariant derivative possesses the essential property that it does not destroy the tensorial character of the object on which it acts: *it generates tensors out of tensors*. Indeed, one can verify that

$$D_\mu \psi_q(x) = \partial_\mu \psi_q(x) - iqA_\mu(x)\psi_q(x) \tag{1.1.16}$$

is once again a tensor of charge q. Another highly useful property of this definition is the validity of the **Leibniz rule** for covariant derivatives: because products of tensors are also tensors, one can systematically check that

$$D_\mu(\psi_{q_1}\psi_{q_2}) = (D_\mu \psi_{q_1})\psi_{q_2} + \psi_{q_1}(D_\mu \psi_{q_2})\,. \tag{1.1.17}$$

1.1.3 | The Electromagnetic Field Tensor and Maxwell Lagrangian

We have demonstrated that local gauge invariance is successfully achieved by introducing the gauge field $A_\mu(x)$, which is readily interpreted physically as the potential of the electromagnetic field. However, having introduced a new dynamical field into the theory, one must grant it suitable dynamics by adding a kinetic term for $A_\mu(x)$ to the lagrangian. This new kinetic term must independently be gauge invariant because the rest of the Lagrangian already is: gauge symmetry remains the guiding principle for building the action.

It is conceptually useful to proceed using our established tensor formalism. We start by calculating the commutator of two covariant derivatives acting on the tensor ψ of charge 1:

$$[D_\mu, D_\nu]\psi = -iF_{\mu\nu}\psi \tag{1.1.18}$$

This relation defines the essential quantity $F_{\mu\nu}$. Since we have strictly tensorial quantities on the left-hand side, the right-hand side must mathematically also be built exclusively out of tensors. In this way, we deduce that $F_{\mu\nu}$ is a tensor of charge $q = 0$, meaning it is a quantity that is strictly invariant under gauge transformations (i.e., it satisfies a transformation with $q = 0$ in (1.1.13)). Computing the left-hand side of (1.1.18) explicitly, one finds

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{1.1.19}$$

which is readily recognized and interpreted as the electromagnetic field tensor. One can manually verify that it is indeed completely gauge invariant, using the transformation rules for the gauge field $A'_\mu(x)$ given in (1.1.11).

Now, it is mathematically straightforward to construct a gauge invariant lagrangian containing at most two derivatives on A_μ. It is sufficient to use as a basic building block the field strength $F_{\mu\nu}$, which being gauge invariant itself makes it incredibly easy to construct further gauge invariant quantities. One must concurrently require Lorentz invariance to maintain a relativistic theory, which ultimately leads to the free Maxwell lagrangian. In a standard normalization, this reads:

$$\mathcal{L}_{Maxwell} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}\,. \tag{1.1.20}$$

1.1.4 | The Complete QED Lagrangian and Feynman Rules

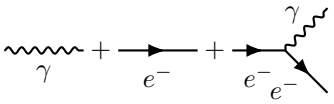
Summing together all the individual pieces that are separately gauge invariant (i.e., combining (1.1.12) and (1.1.20)), one finally finds the full QED lagrangian:

$$\boxed{\mathcal{L}_{QED} = -\frac{1}{4e^2} F_{\mu\nu}F^{\mu\nu} - \bar{\psi}\gamma^\mu D_\mu \psi - m\bar{\psi}\psi} \tag{1.1.21}$$

where a free multiplicative parameter $1/e^2$ is introduced to account for a relative weight between the different terms that are separately gauge invariant.

Let us thoroughly analyze the various terms contained in (1.1.21). It is highly useful to redefine the gauge field via $A_\mu \to eA_\mu$ (doing so recovers the standard normalization of the free Maxwell action) and recognize that e is the physical coupling constant. Note that e now appears inside the covariant derivative $D_\mu = \partial_\mu - ieA_\mu(x)$, taking $e < 0$ for the electron. Expanding this, we obtain:

$$\begin{aligned}
\mathcal{L}_{QED} &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{\psi}(\gamma^\mu D_\mu + m)\psi \\
&= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{\psi}(\gamma^\mu \partial_\mu + m)\psi + ieA_\mu \bar{\psi}\gamma^\mu \psi \\
&= \quad + \quad + \quad
\end{aligned}$$



The first term distinctly describes the free propagation of photons, the second one accounts for the free propagation of electrons, and the crucial third one dictates the elementary interaction between photons and electrons. The constant e is the *coupling*, appropriately identified with the elementary charge of the electron. Thus, the guiding gauge principle has powerfully allowed us to discover the exact interaction mechanism between fields of spin 1/2 and 1. Let us summarize once again the precise rules of gauge transformations for the lagrangian: by rescaling for simplicity also the phase angle $\alpha(x) \to e\alpha(x)$, we have

$$\begin{cases} \psi & \to & \psi' = e^{ie\alpha}\psi \\ \bar{\psi} & \to & \bar{\psi}' = e^{-ie\alpha}\bar{\psi} \\ A_\mu & \to & A'_\mu = A_\mu + \partial_\mu \alpha\,. \end{cases} \tag{1.1.22}$$

1.1.5 | Feynman Diagrams for Basic Scattering Processes

If the fundamental coupling constant e is sufficiently small, the theory can be successfully treated perturbatively, and the theoretical amplitudes for the various physical processes in QED can be systematically associated to Feynman diagrams constructed with the elementary interaction vertex found in (1.1.22). For example, the fundamental electron-electron scattering process (known as Möller scattering) at the lowest perturbative order is represented by:

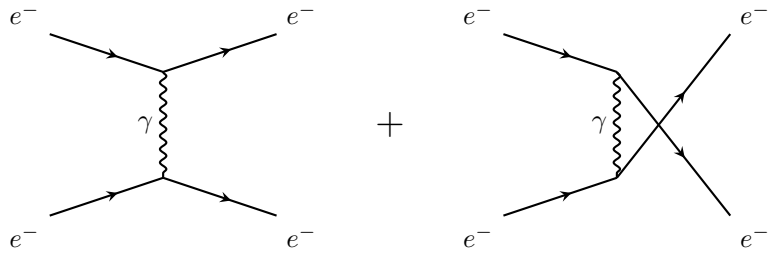


Figure 1.1: Electron-electron scattering (Möller scattering).

where, by our adopted convention, time runs along the horizontal axis in our Feynman diagrams. Other significant processes include electron-positron scattering (Bhabha scattering):

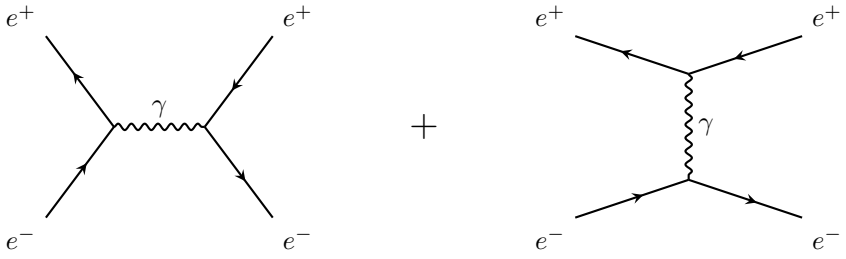


Figure 1.2: Electron-positron scattering (Bhabha scattering).

and the scattering of electrons and photons (Compton scattering):

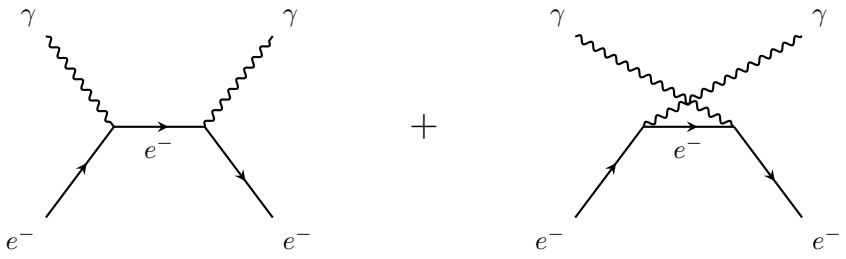


Figure 1.3: Photon-electron scattering (Compton scattering).

Fascinatingly, pure photon-photon scattering is also dynamically possible. Since there is absolutely no elementary vertex coupling four photons directly, the first perturbative term that one calculates is given by the one-loop box graph:

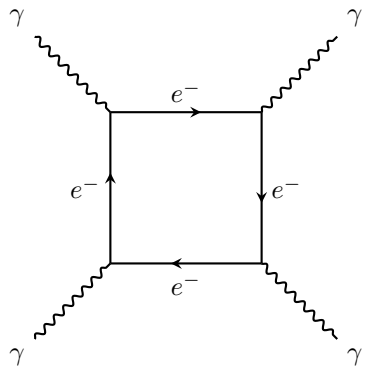


Figure 1.4: Photon-photon scattering (Box diagram).

together with topologically similar graphs where the external photon lines are logically attached to the internal vertices with completely different orderings. In general quantum field theories, loop corrections can be severely divergent and must be rigorously cured by the procedure of renormalization. However, quite remarkably, for photon-photon scattering, the total sum of the one-loop Feynman diagrams indicated above is ultraviolet finite, and strictly no four-photon counterterm is needed in the lagrangian. This elegant finiteness is a direct and beautiful consequence of the underlying gauge invariance of the theory.

1.2 | Non-abelian gauge theories and QCD

The theoretical framework used for the construction of gauge invariant actions can be elegantly extended to accommodate compact non-abelian groups. To accomplish this, we must first review the principal mathematical properties of Lie groups, leaving more detailed foundational considerations for a dedicated appendix.

1.2.1 | Lie groups

Before constructing non-abelian gauge theories, let us briefly review the characteristics of simple and compact Lie groups[2], keeping the special unitary group $SU(N)$ in mind as our primary illustrative example:

$$\mathrm{SU}(N) = \{U \in \mathrm{GL}(N,\mathbb{C}) \,|\, U^\dagger U = 1, \det U = 1\},$$

and we are interested to study $SU(N)$ since we can write the unitary group as a direct product of $U(1)$ and $SU(N)$:

$$U(N) = \cdots = U(1) \times SU(N),$$

which is convenient for our purposes because we have already studied the abelian group $U(1)$ in detail. The group $SU(N)$ is a compact, simple Lie group of dimension $\dim SU(N) = N^2 - 1$.

An arbitrary element U of a compact, non-abelian Lie group G can be parameterized by continuous coordinates α_a (the group parameters), which are formally associated with hermitian generators T^a. A comprehensive list of the main definitions and properties is as follows:

(i) $U = U(\alpha) = \exp(i\alpha_a T^a) \in G \quad a = 1, .., \dim G,$

(ii) $[T^a, T^b] = i f^{ab}{}_c T^c,$

(iii) $\mathrm{tr}(T_F^a T_F^b) = \frac{1}{2}\delta^{ab},$

(iv) $f^{abc} = f^{ab}{}_d \delta^{dc}$ is totally antisymmetric,

(v) $[[T^a, T^b], T^c] + [[T^b, T^c], T^a] + [[T^c, T^a], T^b] = 0 \implies f^{ab}{}_d f^{dc}{}_e + f^{bc}{}_d f^{da}{}_e + f^{ca}{}_d f^{db}{}_e = 0,$

(vi) $(T^a_{\mathrm{Adj}})^b{}_c = -i f^{ab}{}_c.$

Let us elaborate on these defining properties.

Point (i) describes the exponential representation of an arbitrary element U of the group G that is continuously connected to the identity operator. This structure is verified by considering the defining representation, i.e., the specific matrix representation used to define the group. For $SU(N)$, this representation is given by unitary $N \times N$ matrices with a unit determinant. The index a takes as many values as the total dimension of the group. Therefore, an element of the group is parameterized by the continuous "angles" α_a that multiply the generators T^a. Matrices

[2] A *compact group* is defined as a group that is compact as a topological space. Intuitively, it can be conceptualized as a group possessing finite "size" or "volume". More precisely, a compact group has a finite volume with respect to a translation-invariant group theoretical measure known as the "Haar measure". A *simple Lie group* is, roughly speaking, a Lie group whose corresponding Lie algebra $\mathfrak{g}$ is simple. This condition indicates that $\mathfrak{g}$ is non-abelian and possesses no nontrivial ideals. An ideal $\mathfrak{i}$ is a subalgebra satisfying $[\mathfrak{g}, \mathfrak{i}] \subseteq \mathfrak{i}$. Thus, the only ideals of a simple Lie algebra are 0 and $\mathfrak{g}$ itself.

in the defining (also called fundamental) representation are explicitly written as T_F^a, but they are then interpreted as abstract generators T^a that may possess other representations.

Point (ii) provides the **fundamental Lie algebra** satisfied by the hermitian generators T^a. The constants $f^{ab}{}_c$ are strictly real and are known as the **structure constants** of the group. They uniquely characterize the group G and are manifestly antisymmetric in the initial indices a and b.

Point (iii) defines a standard convention for the normalization of the generators in the fundamental representation T_F^a. It formally identifies the so-called "**Killing metric**". More generally, one could have defined the Killing metric γ^{ab} by the trace relation $\mathrm{tr}(T_F^a T_F^b) = \frac{1}{2}\gamma^{ab}$, and rigorously prove that the Killing metric must be positive definite for all compact Lie groups, such as $SU(N)$. The normalized basis chosen above for the generators T_F^a yields the Kronecker delta δ^{ab} as the Killing metric for the compact group G.

Point (iv) utilizes the established Killing metric to systematically raise the lower index of the structure constants. Following this operation, one can show that the resulting symbols f^{abc} are completely antisymmetric: let us prove this.

Proof. Antisymmetry on the indices a and b is intrinsically obvious from property (ii).

Antisymmetry on the indices b and c is systematically deduced by multiplying the Lie algebra with an additional generator, taking the trace, and subsequently using point (iii) in conjunction with the cyclic property of the trace operation:

$$\begin{aligned} f^{abc} &\propto \mathrm{Tr}([T^a, T^b]T^c) = \mathrm{Tr}((T^aT^b - T^bT^a)T^c) = \mathrm{Tr}(T^aT^bT^c) - \mathrm{Tr}(T^bT^aT^c) \\ &= \mathrm{Tr}(T^aT^bT^c) - \mathrm{Tr}(T^aT^cT^b) \qquad \text{(cyclic property)} \\ &= \mathrm{Tr}(T^a(T^bT^c - T^cT^b)) = \mathrm{Tr}(T^a[T^b, T^c]) \\ &= -\mathrm{Tr}(T^a[T^c, T^b]) \propto -f^{acb} \end{aligned}$$

explicitly proving the antisymmetry on the indices b and c. Antisymmetry on the indices a and c is then automatically implied by the previous two results. □

Point (v) explicitly provides the **Jacobi identities** inherently satisfied by the structure constants.

Point (vi) defines the **adjoint representation**. It is formally proven to be a valid mathematical representation by utilizing the Jacobi identities. Note that the indices in $(T^a_{\mathrm{Adj}})^b{}_c$ all run from 1 to $\dim G$, so that the dimension of the adjoint representation naturally coincides with the dimension of the group itself, i.e. $\dim G$. In the specific case of $SU(N)$, these indices run from 1 to $N^2 - 1$, and the dimension of the adjoint representation is thus $N^2 - 1$.

1.2.2 | Action with rigid $SU(N)$ symmetry

Let us now introduce physical fields by considering N free Dirac fields, assembled structurally into column and row vectors as follows:

$$\psi = \begin{pmatrix} \psi^1 \\ \psi^2 \\ . \\ \psi^N \end{pmatrix}, \qquad \bar{\psi} = (\bar{\psi}_1, \bar{\psi}_2, \ldots, \bar{\psi}_N),$$

which let us naturally study the fundamental representation of the group $SU(N)$. The fields are structured so that the corresponding scalar product

$$\bar{\psi}\psi = \bar{\psi}_1\psi^1 + \bar{\psi}_2\psi^2 + \cdots + \bar{\psi}_N\psi^N,$$

is manifestly $SU(N)$ invariant. If these defined fields possess identical masses m, then there exists a global continuous symmetry. The free lagrangian is straightforwardly given by

$$\mathcal{L}_{Dirac} = -\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi, \tag{1.2.1}$$

and is invariant under the rigid $SU(N)$ symmetry transformations provided by

$$\begin{aligned} \psi(x) &\rightarrow \psi'(x) = U\psi(x), \\ \bar{\psi}(x) &\rightarrow \bar{\psi}'(x) = \bar{\psi}(x)U^\dagger = \bar{\psi}(x)U^{-1}, \end{aligned} \tag{1.2.2}$$

where $U \in SU(N)$, and $U^\dagger = U^{-1}$ inherently holds since U is a unitary matrix. These are classified as **global transformations**, as the group parameters α^a contained within $U = U(\alpha) = \exp(i\alpha^a T^a)$ are strictly constant across spacetime (note that indices in α^a are raised and lowered with the Killing metric, which coincides with the identity matrix in our chosen conventions).

It is critical to note that all the fermion masses must be strictly identical; otherwise, the mass term would dynamically break the $SU(N)$ symmetry. There is also an independent abelian $U(1)$ symmetry present, but the procedure for its gauging proceeds exactly as described previously.

1.2.3 | Covariant derivative

To introduce gauge interactions, we would like to mathematically extend the theory to achieve a local $SU(N)$ symmetry. For this purpose, it is once again convenient to introduce the powerful concept of **covariant derivatives**. By formal definition, the covariant derivative successfully produces new tensors when applied to existing tensors. To begin, we recall that a tensor $\psi(x)$ residing in the fundamental representation of the gauge group $SU(N)$, i.e., the representation usually indicated by its dimension $\mathbf{N}$, is a field defined by the transformation:

$$\psi(x) \rightarrow \psi'(x) = U(x)\psi(x),$$

where $U(x)$ is an $N \times N$ matrix of $SU(N)$ that can vary for any spacetime point x. More generally, fields transforming in any given mathematical representation $R(U(x))$ of the original matrices $U(x)$ are formally said to be tensors in the representation R. For example, the conjugate field $\bar{\psi}(x)$ is a tensor in the anti-fundamental representation, which is usually indicated by $\bar{\mathbf{N}}$ as it corresponds to the complex conjugate of the representation $\mathbf{N}$, and its transformation rule is written as follows:

$$\bar{\psi}(x) \rightarrow \bar{\psi}'(x) = \bar{\psi}(x)U(x)^{-1}.$$

Evidently, the bilinear term $\bar{\psi}(x)\psi(x)$ acts as a scalar under this local gauge transformation.

The covariant derivative acting on tensors produces new tensors. It is defined by the following operator:

$$D_\mu = \partial_\mu + W_\mu(x), \tag{1.2.3}$$

where $W_\mu(x)$ is a matrix-valued gauge field, which is also commonly known as the **connection** (geometrically, *it strictly defines a notion of parallel transport in a certain mathematical space*, known formally as a fiber bundle). When applied to the specific tensor $\psi(x)$, the term containing

the gauge field W_μ inherently mixes the N Dirac fermions contained within ψ, and thus W_μ is represented by $N \times N$ matrices (specifically, one matrix for any given value of the spacetime index μ). It performs infinitesimal group transformations and defines a specific sort of parallel transport. It can be conveniently expanded in terms of the group generators as follows:

$$W_\mu(x) = -iW_\mu^a(x)T^a\,. \tag{1.2.4}$$

This relation explicitly defines the anti-hermitian component gauge fields $W_\mu^a(x)$, and there are exactly N^2-1 of them for the gauge group $SU(N)$. From the fundamental physical requirement of covariance:

$$\begin{aligned} \psi(x) \quad &\to \quad \psi'(x) = U(x)\psi(x), \\ D_\mu\psi(x) \quad &\to \quad D'_\mu\psi'(x) = U(x)D_\mu\psi(x), \end{aligned}$$

one systematically obtains the following necessary transformation rule for the gauge potential itself:

$$W_\mu(x) \quad \to \quad W'_\mu(x) = U(x)W_\mu(x)U^{-1}(x) + U(x)\partial_\mu U^{-1}(x)\,. \tag{1.2.5}$$

Let us explicitly prove this relation. By rigorously requiring that $D'_\mu\psi' = UD_\mu\psi$, one computes:

$$\begin{aligned} D'_\mu\psi' &\equiv (\partial_\mu + W'_\mu)\psi' \\ &= UD_\mu\psi = U\partial_\mu\psi + UW_\mu\psi = U\partial_\mu(U^{-1}U\psi) + UW_\mu U^{-1}U\psi \\ &= (U\partial_\mu U^{-1})\psi' + \partial_\mu\psi' + UW_\mu U^{-1}\psi' \\ &= [\partial_\mu + (UW_\mu U^{-1} + U\partial_\mu U^{-1})]\psi', \end{aligned} \tag{1.2.6}$$

which definitively proves the above transformation rule for W_μ. An important clarifying comment: as ψ transforms specifically in the fundamental representation of $SU(N)$, the generators T^a contained within the W_μ of (1.2.4) and (1.2.6) are inherently the generators in the fundamental representation. In general application, one must always use the generators corresponding to the representation of the specific tensors on which the covariant derivative acts.

Covariant derivatives do not commute. This profound geometric fact allows us to systematically define the "curvature" tensor (or physical "field strength") in the following way:

$$[D_\mu, D_\nu]\psi = F_{\mu\nu}\psi, \tag{1.2.7}$$

so that, upon expansion

$$\begin{aligned} [D_\mu, D_\nu]\psi &= (\partial_\mu + W_\mu)(\partial_\nu + W_\nu)\psi - (\partial_\nu + W_\nu)(\partial_\mu + W_\mu)\psi \\ &= (\partial_\mu W_\nu - \partial_\nu W_\mu)\psi + W_\mu\partial_\nu\psi - W_\nu\partial_\mu\psi + W_\mu W_\nu\psi - W_\nu W_\mu\psi \\ &= (\partial_\mu W_\nu - \partial_\nu W_\mu + [W_\mu, W_\nu])\psi, \end{aligned}$$

we find:

$$F_{\mu\nu} = \partial_\mu W_\nu - \partial_\nu W_\mu + [W_\mu, W_\nu]\,. \tag{1.2.8}$$

It is mathematically immediate to check that the resulting object transforms covariantly as:

$$F_{\mu\nu} \quad \to \quad F'_{\mu\nu} = UF_{\mu\nu}U^{-1}, \tag{1.2.9}$$

which follows logically from the foundational covariance of (1.2.7), endowing the field with its tensorial properties. This specific transformation rule corresponds identically to the adjoint representation (the representation of total dimension N^2-1).

1.2.4 | Gauge invariant action

It is now remarkably simple to construct a *globally and locally gauge invariant lagrangian* starting from (1.2.1): it is enough to formally substitute standard derivatives with gauge covariant derivatives (this general procedure is also classically called "*minimal coupling*"). One finds:

$$\mathcal{L}_1 = -\bar{\psi}(\gamma^\mu D_\mu + m)\psi,$$

that dynamically depends on the new physical field W_μ contained inherently in the covariant derivative D_μ. Now, one must necessarily introduce proper dynamics for W_μ as well. By considering the absolute simplest gauge and Lorentz invariant lagrangian term containing at most two derivatives, one is logically led to:

$$\mathcal{L}_2 = \frac{1}{2}\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) = -\frac{1}{4}F^a_{\mu\nu}F^{\mu\nu a}$$

where we have explicitly used the generators in the fundamental representation properly normalized by $\mathrm{tr}\, T^a T^b = \frac{1}{2}\delta^{ab}$. We have chosen here a specific canonical normalization. More generally, one should realistically consider a relative mathematical weight between the different gauge-invariant pieces, so that by formally introducing a physical coupling constant g, one writes the final complete lagrangian as:

$$\boxed{\mathcal{L} = \frac{1}{2g^2}\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) - \bar{\psi}(\gamma^\mu D_\mu + m)\psi} \tag{1.2.10}$$

which is completely gauge invariant under the following comprehensive transformation rules:

$$\begin{aligned}
\psi(x) &\rightarrow \psi'(x) = U(x)\psi(x) \\
\bar{\psi}(x) &\rightarrow \bar{\psi}'(x) = \bar{\psi}(x)U^{-1}(x) \\
W_\mu(x) &\rightarrow W'_\mu(x) = U(x)W_\mu(x)U^{-1}(x) + U(x)\partial_\mu U^{-1}(x)
\end{aligned} \tag{1.2.11}$$

with the very last one logically implying:[3]

$$F_{\mu\nu}(x) \rightarrow F'_{\mu\nu}(x) = U(x)F_{\mu\nu}(x)U^{-1}(x)\,. \tag{1.2.12}$$

Infinitesimal transformation rules. Let us rigorously report the *infinitesimal transformations* as well. By defining the matrix $\alpha \equiv -i\alpha_a T^a$ with Lie parameters $\alpha_a \ll 1$, one concisely describes an infinitesimal transformation in the form:

$$U = e^{i\alpha_a T^a} = e^{-\alpha} = 1 - \alpha + O(\alpha^2) \tag{1.2.13}$$

so that for the corresponding infinitesimal variations of the fields ($\delta\psi(x) \equiv \psi'(x) - \psi(x)$) one systematically finds:

$$\begin{aligned}
\delta\psi &= -\alpha\psi, \\
\delta\bar{\psi} &= \bar{\psi}\alpha, \\
\delta W_\mu &= \partial_\mu\alpha + [W_\mu, \alpha] = D_\mu\alpha, \\
\delta F_{\mu\nu} &= [F_{\mu\nu}, \alpha],
\end{aligned} \tag{1.2.14}$$

where in the last-but-one line, we clearly recognize *the covariant derivative acting specifically on the adjoint representation*. The explicit infinitesimal form of the gauge transformations is fundamentally needed when studying the required gauge-fixing procedure to properly quantize the theory.

[3] In order to check the trasformation rule for the field strenght tensor, one has to explicitly compute all terms, find the correct transformation rule and then use the derivative of the inverse matrix U^{-1} (from chain rule in $\partial_\mu \mathbb{I} = \partial_\mu(UU^{-1}) = 0$) given by $\partial_\mu U^{-1} = -U^{-1}(\partial_\mu U)U^{-1}$ to simplify all the other terms.

Physical Interpretation and Form of the Lagrangian

Now, let us rewrite the total lagrangian by considering a helpful field *redefinition*, implemented by the simple rescaling $W_\mu \to gW_\mu$. It is primarily used to get a canonical kinetic normalization for the gauge field, which practically amounts to pushing the explicit coupling constant directly in front of the interaction vertices. The shift naturally induces a corresponding *redefinition* of the field strength $F_{\mu\nu} \to gF_{\mu\nu}$, with the new modified field strength given by:

$$F_{\mu\nu} = \partial_\mu W_\nu - \partial_\nu W_\mu + g[W_\mu, W_\nu]\,. \tag{1.2.15}$$

In terms of their isolated components,

$$\begin{aligned} W_\mu(x) &= -iW^a_\mu(x)T^a, \\ F_{\mu\nu}(x) &= -iF^a_{\mu\nu}(x)T^a, \end{aligned} \tag{1.2.16}$$

and

$$F^a_{\mu\nu} = \partial_\mu W^a_\nu - \partial_\nu W^a_\mu + g f^{abc} W^b_\mu W^c_\nu, \tag{1.2.17}$$

and the total lagrangian ultimately takes the more explicit physical form:

$$\boxed{\mathcal{L} = -\frac{1}{4}F^a_{\mu\nu}F^{\mu\nu a} - \bar{\psi}[\gamma^\mu(\partial_\mu - igW^a_\mu T^a) + m]\psi}\,, \tag{1.2.18}$$

where the contraction of the group indices a is always implicitly understood as a trace over the group indices.

The very first term in the explicit action(1.2.18) elegantly describes the free space propagation of the fields W^a_μ (the fundamental non-abelian spin 1 particle) along with associated cubic and quartic self-interactions inherently governed by coupling constants g and g^2, respectively. A non positive-definite Killing metric would drastically result in a term with kinetic energy that is definitively not positive-definite, and this would structurally not be acceptable: it is physically necessary to consider only compact groups to successfully satisfy this stability request. The crucial second term describes the free propagation of the ψ fields (spin 1/2 particles endowed with non-abelian charges, i.e., "color" charges) working intimately together with an elementary interaction with the gauge field W^a_μ possessing a strength dynamically measured by the coupling constant g. It can be effectively treated perturbatively if g is sufficiently small enough. The conceptual "non-abelian" or "color" charge strictly corresponds to the mathematical representation of the gauge group chosen for the matter fields ψ (in our specific case we have cleanly taken the fundamental representation, but logically any other representation could have been validly chosen as well). Thus, we have explicitly seen how the fundamental principle of local gauge invariance has powerfully allowed us to systematically derive all the interaction vertices between diverse fields of spin 1/2 and 1 directly in terms of the single universal coupling constant g.

The physical coupling constant. The physical coupling constant now visibly appears directly in front of the fundamental interaction vertices. The infinitesimal gauge transformations, after systematically redefining the parameters $\alpha^a \to g\alpha^a$, read explicitly:

$$\begin{aligned} \delta\psi(x) &= ig\alpha^a(x)T^a\psi(x) \\ \delta\bar{\psi}(x) &= -ig\alpha^a(x)\bar{\psi}(x)T^a \\ \delta W^a_\mu(x) &= \partial_\mu\alpha^a(x) + gf^{abc}W^b_\mu(x)\alpha^c(x) = D_\mu\alpha^a(x) \end{aligned} \tag{1.2.19}$$

where T^a are, as always, the generators in the fundamental representation.

Algebraic Structure of the Infinitesimal Transformations

It is highly useful to analytically examine in more precise detail the algebraic structure of the infinitesimal transformation laws that we have been rigorously deriving. Introducing explicit matrix indices, the infinitesimal transformation of the fermion field $\psi(x)$ from (1.2.19) is formally written as:

$$\delta\psi^i(x) = ig\alpha^a(x)(T_F^a)^i{}_j\psi^j(x) \tag{1.2.20}$$

with indices $i, j = 1, ..., N$, and where by $(T_F^a)^i{}_j$ we now clearly indicate the individual matrix elements of the generators in the fundamental representation ($N \times N$ matrices, exactly one for each value of the group index a). Similarly, the transformation of the Dirac conjugate field (morally acting as the complex conjugate field[4]) is rigorously written as:

$$\delta\bar{\psi}_i(x) = -ig\alpha^a(x)(T_F^{a*})_i{}^j\bar{\psi}_j(x) = ig\alpha^a(x)(T_{\bar{F}}^a)_i{}^j\bar{\psi}_j(x)\,, \tag{1.2.21}$$

where $T_{\bar{F}}^a = -T_F^{a*}$ strictly are the generators residing in the complex conjugate of the fundamental representation T_F^a. Also, we may meticulously check that the non-abelian field strength $F_{\mu\nu}^a$ correctly transforms strictly in the adjoint representation:

$$\delta F_{\mu\nu}^a = gf^{abc}F_{\mu\nu}^b\alpha^c = ig\alpha^c(T_{\text{Adj}}^c)^{ab}F_{\mu\nu}^b\,, \tag{1.2.22}$$

precisely as the generators in the defined adjoint representation are rigorously given by:

$$(T_{\text{Adj}}^c)^{ab} = -if^{abc}\,. \tag{1.2.23}$$

Let us also verify explicitly and algebraically that the established transformation law of W_μ^a can be completely expressed exactly in terms of the covariant derivative acting dynamically on a tensor situated in the adjoint representation:

$$\delta W_\mu^a = \partial_\mu\alpha^a + gf^{abc}W_\mu^b\alpha^c = \partial_\mu\alpha^a - igW_\mu^b(T_{\text{Adj}}^b)^{ac}\alpha^c = D_\mu\alpha^a\,. \tag{1.2.24}$$

Note critically also that the isolated non-derivative part of this specific transformation can be formally written equivalently as:

$$\delta W_\mu^a = ig\alpha^c(T_{\text{Adj}}^c)^{ab}W_\mu^b + \cdots\,, \tag{1.2.25}$$

which prominently highlights the underlying tensorial character of this specific part of the transformation, perfectly matching (1.2.22).

Finally, let us formally recall that the continuous Jacobi identities for completely arbitrary operators, once carefully applied directly to the covariant derivative operator $D_\mu = \partial_\mu + gW_\mu$

$$[D_\mu, [D_\nu, D_\lambda]] + [D_\nu, [D_\lambda, D_\mu]] + [D_\lambda, [D_\mu, D_\nu]] = 0\,,$$

give geometric rise directly to the so-called mathematical **Bianchi identities** strictly governing the field strength $F_{\mu\nu}$:

$$D_\mu F_{\nu\lambda} + D_\nu F_{\lambda\mu} + D_\lambda F_{\mu\nu} = 0\,, \tag{1.2.26}$$

where the relevant covariant derivative evaluated in the adjoint representation is written as:

$$D_\mu F_{\nu\lambda} = \partial_\mu F_{\nu\lambda} + g[W_\mu, F_{\nu\lambda}],$$

that, when broken down in explicit components, structurally takes the complete form:

$$D_\mu F_{\nu\lambda}^a = \partial_\mu F_{\nu\lambda}^a + gf^{abc}W_\mu^b F_{\nu\lambda}^c\,.$$

[4]We logically recall here that upper dotted indices are mathematically equivalent to lower undotted indices for the defining $SU(N)$ group.

1.2.5 | The action of quantum chromodynamics (QCD)

The Gauge Group and Gluons

The fundamental action of quantum chromodynamics (QCD) is built upon the non-abelian gauge group $SU(3)$. In addition to the gluons (the eight spin-1 particles associated with the gauge field W_μ^a, which inherently possess an index in the adjoint representation and thus belong to the **8** representation of $SU(3)$) the total lagrangian contains six distinct fermion fields ψ_f corresponding to the six known flavors of quarks, denoted by $f = (u, d, c, s, t, b)$. These correspond to the up, down, charm, strange, top, and bottom quarks, respectively.

Each individual quark flavor is degenerate in the sense that it transforms in the fundamental **3** representation of the $SU(3)$ gauge group: the quark is physically said to be "colored" (typically denoted with colors red, green, and blue, following the usual convention). The absence of color geometrically indicates a scalar under the gauge group, much like the total lagrangian itself (a scalar quantity rigorously corresponds to the trivial representation **1** of $SU(3)$).

Naturally, the corresponding antiparticles, the antiquarks $(\bar{u}, \bar{d}, \bar{c}, \bar{s}, \bar{t}, \bar{b})$, transform in the complex conjugate representation, the $\bar{\mathbf{3}}$ of $SU(3)$, which mathematically corresponds to the representation of $\bar{\psi}_f$, the Dirac conjugate of ψ_f[5].

Gell-Mann Matrices and the QCD Lagrangian

The eight infinitesimal generators of $SU(3)$ evaluated in the fundamental representation are explicitly given by the Gell-Mann matrices λ^a, which systematically generalize the well-known Pauli matrices σ^i of $SU(2)$ to the higher-dimensional $SU(3)$ case:

$$T^a = \frac{\lambda^a}{2} \qquad a = 1, \dots, 8 \tag{1.2.27}$$

where these matrices are defined as:

$$\begin{aligned}
\lambda^1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
\lambda^4 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \lambda^5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} \\
\lambda^6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \lambda^7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \qquad \lambda^8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.
\end{aligned} \tag{1.2.28}$$

These specific generators are mathematically normalized so that they satisfy $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$.

An arbitrary element of the $SU(3)$ group residing in the fundamental representation is therefore formally given by 3×3 matrices taking the form $U = \exp\left(i\alpha_a \frac{\lambda^a}{2}\right)$. By explicitly calculating the Lie algebra relations $[T^a, T^b] = i f^{abc} T^c$, one systematically finds the explicit values of the

[5] Given the defining matrices in the fundamental representation $U = e^{i\alpha^a T_F^a}$, the complex conjugate matrices also elegantly furnish a valid representation $U^* = e^{-i\alpha^a T_F^{a*}} = e^{i\alpha^a(-T_F^{a*})} \equiv e^{i\alpha^a T_{\bar{F}}^a}$ so that $T_{\bar{F}}^a = -T_F^{a*} = -T_F^{aT}$, where the last relation follows directly because the established generators are hermitian.

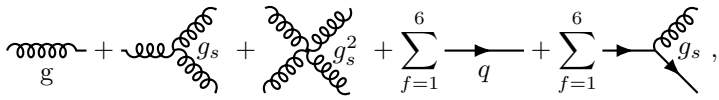
structure constants f^{abc} belonging to the $SU(3)$ group. They are totally antisymmetric under any permutation of indices and are given by:

$$f^{123} = 1$$
$$f^{147} = -f^{156} = f^{246} = f^{257} = f^{345} = -f^{367} = \frac{1}{2}$$
$$f^{458} = f^{678} = \frac{\sqrt{3}}{2}$$

while all other f^{abc} strictly not related to these by permuting the indices are identically zero. As an instructive exercise, one may try to explicitly verify some of these numbers.

The complete **QCD lagrangian** is therefore constructed as:

$$\begin{aligned}\mathcal{L}_{QCD} &= -\frac{1}{4}F^a_{\mu\nu}F^{\mu\nu a} - \sum_{f=1}^{6}\bar{\psi}_f\Big(\gamma^\mu D_\mu + m_f\Big)\psi_f \\ &= -\frac{1}{4}F^a_{\mu\nu}F^{\mu\nu a} - \sum_{f=1}^{6}\bar{\psi}_f\Big(\gamma^\mu \partial_\mu + m_f\Big)\psi_f + i\frac{g_s}{2}W^a_\mu \sum_{f=1}^{6}\bar{\psi}_f\gamma^\mu\lambda^a\psi_f \qquad (1.2.29)\end{aligned}$$



where the strong coupling constant is conventionally denoted by g_s, and the sum index $f \in (1, 2, \ldots, 6) = (u, d, c, s, t, b)$ specifically indicates the distinct flavor of the quark. Different flavors of quarks physically possess different masses m_f. Note carefully that to obtain the valid mathematical propagator of the gauge field from the very first term, as indicated conceptually in the corresponding figure, one must first rigorously implement a formal gauge-fixing procedure.

Flavor Symmetries and Symmetries of the Action

The QCD lagrangian dynamically possesses additional rigid global symmetries. A well-known rigid symmetry is the $U(1)$ symmetry which uniformly rotates all fields of the quarks by the exact same overall phase ($\psi_f \to \psi'_f = e^{i\alpha/3}\psi_f$ for all f, where the factor of $\frac{1}{3}$ is strictly conventional): the associated conserved physical charge is the *baryon number* B. It is a robust symmetry that is faithfully preserved also by the other fundamental interactions evaluated at the perturbative level (nonperturbative electroweak transitions dynamically violate $B+L$ while strictly preserving $B-L$ with L denoting the lepton number).

Other independent $U(1)$ symmetries rotate the various fermionic fields entirely separately. They logically give rise to precise conservation laws of the respective *fermion numbers* (e.g. strangeness $\mathcal{S}$, charm $\mathcal{C}$, etc.). These distinct flavor symmetries are exactly preserved for QCD (and QED), but the weak force structurally violates them. There are exactly six $U(1)$ independent conserved charges, one for each specific quark flavor, and the global baryon number B is mathematically a particular linear combination of these six independent initial charges. Also, the physical *electric charge* Q is a specific linear combination of them, and it is precisely the one which is eventually locally gauged to correctly obtain the electromagnetic couplings.

A comprehensive summary of these $U(1)$ symmetries is explicitly given in the following table, which rigorously reports the various $U(1)$ charges with a widely accepted standard normalization:

Quarks	$\mathcal{U}$	$\mathcal{D}$	$\mathcal{C}$	$\mathcal{S}$	$\mathcal{T}$	$\mathcal{B}$	B	Q
u	1	0	0	0	0	0	$\frac{1}{3}$	$\frac{2}{3}$
d	0	-1	0	0	0	0	$\frac{1}{3}$	$-\frac{1}{3}$
c	0	0	1	0	0	0	$\frac{1}{3}$	$\frac{2}{3}$
s	0	0	0	-1	0	0	$\frac{1}{3}$	$-\frac{1}{3}$
t	0	0	0	0	1	0	$\frac{1}{3}$	$\frac{2}{3}$
b	0	0	0	0	0	-1	$\frac{1}{3}$	$-\frac{1}{3}$

Table 1.1: Summary of the $U(1)$ charges for the six quark flavors: the *baryon number* B and the *electric charge* Q are specific linear combinations of the six independent *flavor charges*. Note critically that we have indicated the baryon number explicitly by B, and the bottom (or beauty) quantum number by $\mathcal{B}$.

For each continuous symmetry, each quark flavor formally transforms with the specific charge indicated in the table. For example, for the physical electric charge Q we have the transformation:

$$\psi_f \to \psi'_f = e^{i\alpha Q_f}\psi_f \,. \tag{1.2.30}$$

By carefully looking at the generated table, one immediately recognizes the following algebraic relations:

$$\begin{aligned} B &= \frac{1}{3}(\mathcal{U}+\mathcal{C}+\mathcal{T}) - \frac{1}{3}(\mathcal{D}+\mathcal{S}+\mathcal{B}) \\ Q &= \frac{2}{3}(\mathcal{U}+\mathcal{C}+\mathcal{T}) + \frac{1}{3}(\mathcal{D}+\mathcal{S}+\mathcal{B})\,. \end{aligned} \tag{1.2.31}$$

Isospin and the Eightfold Way

There are also highly useful approximate symmetries characterizing the QCD lagrangian. In the theoretical limit in which some of the quark masses are mathematically taken to be strictly identical, there emerges a rigid additional non-abelian symmetry. For example, assuming identically matched masses for the up and down quarks, $m_u = m_d$, one can continuously mix the specific fields ψ_u and ψ_d with a $SU(2)$ matrix to effectively redefine what is physically meant by "up" and "down":

$$\begin{pmatrix}\psi_u \\ \psi_d\end{pmatrix} \to \begin{pmatrix}\psi'_u \\ \psi'_d\end{pmatrix} = U \begin{pmatrix}\psi_u \\ \psi_d\end{pmatrix} \qquad U \in SU(2)\,. \tag{1.2.32}$$

This rigid $SU(2)$ symmetry directly corresponds to the **strong isospin** $\vec{I}$, successfully used to properly group hadrons into distinct families or multiplets. From physical phenomenology, one knows that the strong force tightly binds states of quarks together and clearly shows the profound phenomenon of *color confinement*: physically bound states of quarks are always color singlets and directly correspond to the observable mesons and baryons. Key examples of these families are: (i) the isospin doublet of the core nucleons (proton and neutron) structurally composed of three confined up and down quarks; (ii) the triplet of π mesons, the pions $\pi^{\pm}$ and π^0, fundamentally composed of a quark and an antiquark entirely of the up and down types.

Considering structurally identical masses for the quarks up, down, and strange, $m_u = m_d = m_s$, one logically finds an *even larger symmetry group*, the $SU(3)$ flavor group, that continuously mixes the three flavors up, down, and strange:

$$\begin{pmatrix} \psi_u \\ \psi_d \\ \psi_s \end{pmatrix} \to \begin{pmatrix} \psi'_u \\ \psi'_d \\ \psi'_s \end{pmatrix} = U \begin{pmatrix} \psi_u \\ \psi_d \\ \psi_s \end{pmatrix} \qquad U \in SU(3)\,. \tag{1.2.33}$$

This $SU(3)$ flavor group is precisely the one that is heavily used in the static quark model (the famous "**eightfold way**" of Gell-Mann) to neatly take care of the numerous similarities observationally observed between the various hadrons. It should strictly not be confused with the foundational color group, which is dynamically also an $SU(3)$ group. As already emphatically said, color is expected to tightly confine inside the hadrons and thus leaves only composite colorless states physically observable. Key examples of observable multiplets of hadronic particles successfully described by the $SU(3)$ flavor group are:

- the meson octet $(\pi^{\pm}, \pi^0, K^{\pm}, K^0, \bar{K}^0, \eta_8)$,
- the baryon octet $(p, n, \Sigma^{\pm}, \Sigma^0, \Xi^0, \Xi^-, \Lambda)$,
- the baryon decuplet $(\Delta^-, \Delta^0, \Delta^+, \Delta^{++}, \Sigma^{*\pm}, \Sigma^{*0}, \Xi^{*0}, \Xi^{*-}, \Omega^-)$.

The physical existence of these structured families is entirely compatible with formal group theory: the **8** and the **10** are well-known representations of $SU(3)$. Let us logically consider the mesons in more precise detail. They phenomenologically consist of a quark-antiquark pair $(q\bar{q})$. The quarks q transform dynamically in the **3** of $SU(3)$, with $\mathbf{3} \sim (u, d, s)$, while the antiquarks $\bar{q}$ transform mathematically in the $\bar{\mathbf{3}}$ of $SU(3)$, with $\bar{\mathbf{3}} \sim (\bar{u}, \bar{d}, \bar{s})$. From this, it logically follows that possible physically bound states $(q\bar{q})$ must transform in the tensor product space:

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}$$

and therefore both singlet and octets could, in theoretical principle, exist physically for the mesons.

Baryons are tightly bound states of three quarks (qqq), and since $SU(3)$ group theory mathematically tells us that

$$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = (\mathbf{6} \oplus \bar{\mathbf{3}}) \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}$$

we intuitively understand that octets and decuplets are formally allowed structural possibilities for multiplets of baryons.

It is also theoretically useful to consider a noticeably simpler version of QCD strictly containing only three massless flavours of fermions. This can often provide a highly good approximation, since the actual physical masses of the three lightest quarks can often be reasonably neglected in many high-energy regimes. In this idealized limit, the full global internal symmetry is:

$$U(3)_L \times U(3)_R = U(3)_V \times U(3)_A = U(1)_V \times SU(3)_V \times U(1)_A \times SU(3)_A\,. \tag{1.2.34}$$

The axial symmetry $SU(3)_A$ is dynamically spontaneously broken, logically giving rise to eight purely massless Goldstone bosons, which are phenomenologically identified with the meson octet explicitly described above. By contrast, the $U(1)_A$ symmetry is fundamentally broken by the quantum axial anomaly and therefore does not survive as a true symmetry at the fully quantum level. Consequently, strictly no corresponding physical Goldstone boson is expected.

1.3 | Quantization of Gauge Theories

The formulation of fundamental interactions through gauge theories presents a profound conceptual challenge when transitioning from classical physics to a consistent quantum field theory. The root of this difficulty lies in **local gauge invariance**: rather than representing a standard physical symmetry between distinct physical states, gauge invariance signifies an intrinsic mathematical redundancy. Specifically, while a massless spin-1 field propagates only two physical transverse polarization degrees of freedom, manifest Lorentz covariance and spacetime locality require describing the interaction through a four-vector gauge potential A_μ. Reconciling this redundancy with standard quantization schemes is notoriously non-trivial.

In canonical Hamiltonian quantization, this feature manifests as a constrained dynamical system. Because the classical lagrangian contains no time derivative of the temporal component A_0, the conjugate momentum π^0 vanishes identically, leading to primary and secondary constraints (such as Gauss's law) in Dirac's constraint classification. Solving these constraints explicitly prior to quantization spoils manifest Lorentz invariance. Conversely, quantizing first in an enlarged Hilbert space with an indefinite metric and selecting physical states via operator constraints—as in the Gupta–Bleuler method for abelian Quantum Electrodynamics (QED)—fails to generalize cleanly to non-abelian Yang–Mills theories, where non-linear self-couplings prevent the straightforward decoupling of unphysical longitudinal and timelike states.

The Feynman path integral approach provides a natural, manifestly covariant framework, yet the gauge redundancy resurfaces as a severe divergence. The generating functional integrates the phase factor $e^{iS[A]}$ over all field configurations:

$$Z = \int \mathcal{D}A\, e^{iS[A]}\,.$$

Because the classical action $S[A]$ is constant along continuous, infinite-dimensional **gauge orbits** (sets of gauge-equivalent configurations), the integration redundantly sums over identical physical states. This multiplies the path integral by the infinite volume of the gauge group, $\mathrm{Vol(Gauge)} \to \infty$, rendering Z ill-defined. Perturbatively, this pathology reflects the non-invertibility of the quadratic kinetic operator for A_μ, whose zero modes along longitudinal directions prevent the construction of a well-defined free gauge-field propagator.

To obtain a predictive and finite quantum theory, one must constrain the functional domain to intersect each gauge orbit exactly once, a procedure known as **gauge-fixing**. The systematic implementation of this constraint is achieved via the **Faddeev–Popov method**: by inserting a functional resolution of identity into the path integral, the infinite gauge volume is factorized out, and the resulting functional determinant (the Faddeev–Popov determinant) is exponentiated through auxiliary, anticommuting scalar fields termed *ghosts*.

Crucially, while the gauge-fixing condition explicitly breaks local gauge invariance, the resulting effective quantum action retains a fundamental, rigid graded symmetry: the **BRST (Becchi–Rouet–Stora–Tyutin) symmetry**. As developed in the following sections, the nilpotency of the BRST charge and the corresponding cohomology framework ensure that physical states, scattering matrix elements, and quantum unitarity remain strictly independent of the chosen gauge-fixing condition.

1.3.1 | Faddeev Popov Quantization

The quantization of gauge theories necessitates special mathematical considerations to systematically construct a well-defined Quantum Field Theory (QFT) and its corresponding perturbative expansion. The fundamental problem becomes immediately clear within the path integral approach, where a naive path integral quantization yields a severely diverging result. Let us consider the illustrative case of the abelian $U(1)$ theory, which describes free electromagnetism.

The classical action and the naive path integral are given by:

$$\begin{aligned} S[A] &= \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu}\right), \\ Z &= \int \mathcal{D}A\, e^{iS[A]} \to \infty, \end{aligned} \tag{1.3.1}$$

where the electromagnetic field strength tensor is defined as $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Here, the path integral inherently diverges because one is redundantly summing over an infinite number of gauge-equivalent field configurations. Specifically, configurations related by the local gauge transformation

$$A_\mu(x) \to A^g_\mu(x) = A_\mu(x) + ig(x)\partial_\mu g^{-1}(x)\,, \qquad g(x) = e^{i\alpha(x)} \in U(1), \tag{1.3.2}$$

possess the exact same value of the action, $S[A^g] = S[A]$. Consequently, the entire field space dynamically decomposes into inequivalent continuous **gauge orbits**, as visualized conceptually in the figure below.

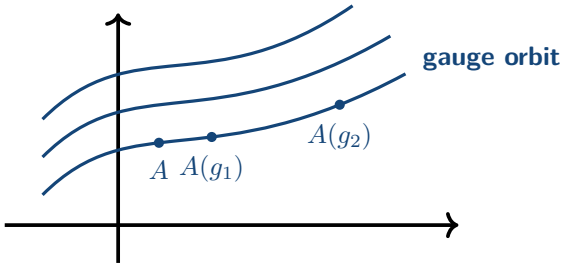


Figure 1.5: Gauge orbits in field space. The field configurations A, $A(g_1)$, and $A(g_2)$ lie on the same gauge orbit, meaning they are physically equivalent and connected by a gauge transformation $g(x) \in U(1)$.

One would ideally like to formally define the path integral in such a way as to obtain a finite and explicitly gauge-invariant result by factoring out this redundant infinite volume:

$$Z = \int \frac{\mathcal{D}A}{\mathrm{Vol(Gauge)}} e^{iS[A]} \sim \text{finite}, \tag{1.3.3}$$

where the formal quantity "Vol(Gauge)" indicates the infinite volume of the gauge group. This conceptual definition can be implemented concretely using a *gauge-fixing function à la Faddeev-Popov*, where unphysical "ghost" fields are elegantly introduced to exponentiate a crucial measure factor. Ideally, the chosen gauge-fixing function should pick exactly one single representative from each gauge orbit, as sketched in figure 1.6. This idealized situation precisely corresponds to the choices of the so-called "**unitary gauges**". However, for practical perturbative calculations, it is often enough to merely reduce the number of representatives for each gauge orbit, which naturally happens in the standard choices of **covariant gauges**.

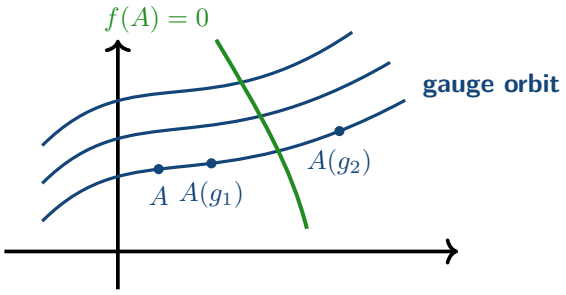


Figure 1.6: Gauge fixing by the condition $f(A) = 0$. The gauge-fixing function intersects the field space such that it picks ideally just one representative from each gauge orbit.

As we shall explore, the gauge-fixed action thus obtained using covariant gauges exhibits a powerful residual global symmetry, called BRST symmetry. This symmetry is a direct remnant of the original gauge symmetry and is deeply instrumental in rigorously proving many properties of quantum gauge theories. In particular, it can be utilized to formally prove the independence of physical, observable quantities from the arbitrary choice of gauge-fixing function. It also forms the foundational basis of the BRST lagrangian quantization method, which systematically generalizes the Faddeev-Popov method to more complex, general cases. An even more powerful advanced method utilizes additional fields (the so-called antifields) and is widely known as the Batalin-Vilkovisky method.

The Faddeev-Popov Trick

Let us start by explicitly describing the Faddeev-Popov method in the simpler free abelian gauge theory case. The clever trick is to use a gauge-fixing condition, as depicted in figure 1.6, and insert the identity (written in a suitable integral way) directly into the path integral to effectively extract the volume of the gauge group. Consider the elementary 1-dimensional identity written as:

$$1 = \int df\, \delta(f) = \int \frac{\partial f(y)}{\partial y} dy\, \delta(f(y)) = \int dy\, \delta(f(y)) \frac{\partial f(y)}{\partial y}$$

which smoothly generalizes to n-dimensions where f becomes a vector f^i with n components:

$$1 = \int d^n f\, \delta^{(n)}(f) = \int d^n y\, \delta^{(n)}(f(y)) \det\left(\frac{\partial f^i(y)}{\partial y^j}\right) .$$

We can formally generalize this finite-dimensional expression further to functional integrals over continuous fields:

$$1 = \int \mathcal{D}g\, \delta\Big(f(A^g(x))\Big) \mathrm{Det}\left(\frac{\delta f(A^g(x))}{\delta g(y)}\right) \tag{1.3.4}$$

where $\mathcal{D}g$ is a suitable gauge-invariant measure that also fundamentally produces the volume of the gauge group, $\int \mathcal{D}g = \mathrm{Vol(Gauge)}$. The notation "Det" strictly indicates a functional determinant. The delta function (which is actually a delta *functional*) $\delta(f(x))$ mathematically means that the whole function $f(x)$ is strictly set to vanish.

Then, let us compute formally the path integral defined in (1.3.3) by systematically inserting the

functional identity from (1.3.4):

$$\begin{aligned} Z &= \int \frac{\mathcal{D}A}{\mathrm{Vol(Gauge)}} e^{iS[A]} \\ &= \int \frac{\mathcal{D}A}{\mathrm{Vol(Gauge)}} \int \mathcal{D}g\, \delta\Big(f(A^g(x))\Big) \mathrm{Det}\left(\frac{\delta f(A^g(x))}{\delta g(y)}\right) e^{iS[A]} \\ &= \int \frac{\mathcal{D}A^g \mathcal{D}g}{\mathrm{Vol(Gauge)}} \,\delta\Big(f(A^g(x))\Big) \mathrm{Det}\left(\frac{\delta f(A^g(x))}{\delta g(y)}\right) e^{iS[A^g]} \\ &= \int \frac{\mathcal{D}g}{\mathrm{Vol(Gauge)}} \int \mathcal{D}A\, \delta\Big(f(A(x))\Big) \mathrm{Det}\left(\frac{\delta f(A^g(x))}{\delta g(y)}\right) \Bigg|_{g=1} e^{iS[A]} \\ &= \int \mathcal{D}A\, \delta\Big(f(A(x))\Big) \mathrm{Det}\left(\frac{\delta f(A^g(x))}{\delta g(y)}\right) \Bigg|_{g=1} e^{iS[A]} \end{aligned}$$

which is precisely the gauge-fixed path integral we were actively looking for. In these algebraic manipulations, we have first inserted the identity (1.3.4), and then utilized the fundamental fact that the action and the measure are both completely gauge invariant, namely $S[A^g] = S[A]$ and $\mathcal{D}A^g = \mathcal{D}A$. This is certainly mathematically true for the classical action, but it is an implicit assumption for the measure (more concretely, it is intimately related to the regularization methods used to make strict sense of the diverging Feynman diagrams characterizing the perturbative expansion[6]). Then, we have formally changed integration variables from A^g to A, so that nothing in the integrand depends on the gauge parameter $g(x)$ anymore. Consequently, the integration on $\mathcal{D}g$ can be cleanly factorized out to exactly cancel the infinite gauge volume Vol(Gauge)[7].

The final formula is theoretically correct but can be elegantly written in a significantly more useful form by performing two essential steps:

(i) introducing **ghost fields**, i.e., anticommuting fields $c(x)$ and $\bar{c}(x)$ to exponentiate the functional determinant, broadly known as the Faddeev-Popov determinant,

(ii) modifying the gauge-fixing function to effectively get rid of the rigid delta functional from the path integral: instead of strictly setting $f(A(x)) = 0$, one may equivalently set it to $f(A(x)) = h(x)$ (where $h(x)$ is an arbitrary continuous function) and then functionally average over the diverse functions $h(x)$ with a gaussian weight $e^{-\frac{1}{2\xi}\int h^2}$.

The rigorous physical rationale behind this averaging procedure is that inherently different gauge-fixing functions must ultimately give the exact same gauge-invariant result. In particular, nothing physical should depend on the specific numerical value of the gauge parameter ξ.

Thus, implementing these exponentiation steps, one initially finds:

$$\begin{aligned} Z = &\int \mathcal{D}A\, \mathcal{D}c\, \mathcal{D}\bar{c}\, \mathcal{D}h\, \delta\Big(f(A(x)) - h(x)\Big) \\ &\times \exp i\left(S[A] + \int d^4x d^4y\, \bar{c}(x) \frac{\delta f(A^g(x))}{\delta g(y)}\Bigg|_{g=1} c(y) - \frac{1}{2\xi}\int d^4x\, h^2(x)\right), \end{aligned}$$

which is drastically simplified by path integrating over the auxiliary function $h(x)$ to cleanly eliminate the delta functional, allowing us to find in the exponent the complete gauge-fixed total

[6] In chiral theories where the chosen regularization might fail to actively maintain gauge invariance, one physically speaks of **anomalies**. The entire quantization procedure breaks down if the potential anomalies in the gauge symmetry are not exactly canceled.

[7] The formal mathematical identification $\delta g \sim g^{-1} dg$ yields a strictly group-invariant measure, the so-called Haar measure.

action S_{tot}:

$$\begin{aligned} Z &= \int \mathcal{D}A\mathcal{D}c\mathcal{D}\bar{c}\, \exp i \left(S[A] + \int d^4x d^4y\, \bar{c}(x) \frac{\delta f(A^g(x))}{\delta g(y)}\bigg|_{g=1} c(y) - \frac{1}{2\xi}\int d^4x\, f^2(A(x)) \right) \\ &= \int \mathcal{D}A\mathcal{D}c\mathcal{D}\bar{c}\, e^{iS_{\text{tot}}[A,c,\bar{c}]}\,, \end{aligned} \tag{1.3.5}$$

where the action appearing in the exponent has to be interpreted as the *complete gauge-fixed action*.

Example (The Lorenz Gauge). To exemplify the abstract construction derived above, let us explicitly choose a specific gauge-fixing function:

$$f(A) = \partial^\mu A_\mu, \tag{1.3.6}$$

which corresponds directly to the strict *Lorenz gauge* for the initial path integral derivation, and to a weighted Lorenz gauge for the path integral in (1.3.5) (also frequently called the R_ξ gauge). Under an infinitesimal gauge variation $\delta A_\mu = \partial_\mu \alpha$, the gauge-fixing function varies algebraically as:

$$\delta f(A) = \partial^\mu \delta A_\mu = \partial^\mu \partial_\mu \alpha,$$

and one is mathematically led to naturally interpret the functional derivative operator as:

$$\frac{\delta f(A^g(x))}{\delta g(y)}\bigg|_{g=1} \sim \frac{\delta f(A(x))}{\delta \alpha(y)} = \partial^\mu \partial_\mu \delta^4(x-y),$$

whose functional determinant (the Faddeev-Popov determinant) can be faithfully reproduced by path integrating over the anti-commuting scalar ghost fields $c(x)$ and $\bar{c}(x)$. Thus, the complete gauge-fixed action reads:

$$S_{\text{tot}}[A,c,\bar{c}] = S[A] + \int d^4x \left(-\partial^\mu \bar{c}\partial_\mu c - \frac{1}{2\xi}(\partial^\mu A_\mu)^2 \right), \tag{1.3.7}$$

which is also called the *quantum action* of the theory.

The corresponding total gauge-fixed lagrangian density is given by:

$$\mathcal{L}_{\text{tot}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \partial^\mu \bar{c}\partial_\mu c - \frac{1}{2\xi}(\partial^\mu A_\mu)^2, \tag{1.3.8}$$

which, up to total derivative terms that are usually dropped in the action since they do not affect equations of motion, may be symmetrically written as:

$$\mathcal{L}_{\text{tot}} = -\frac{1}{2}\partial^\mu A^\nu \partial_\mu A_\nu + \frac{1}{2}\left(1 - \frac{1}{\xi}\right)(\partial^\mu A_\mu)^2 - \partial^\mu \bar{c}\partial_\mu c\,. \tag{1.3.9}$$

In particular, the highly convenient Feynman gauge is obtained by setting $\xi = 1$ and yields the remarkably simple action:

$$\mathcal{L}_{\text{tot}} = -\frac{1}{2}\partial^\mu A^\nu \partial_\mu A_\nu - \partial^\mu \bar{c}\partial_\mu c\,. \tag{1.3.10}$$

Because the path integral is now perfectly well-defined, one can safely add external sources and compute the fundamental propagators of the theory. In the Feynman gauge, they read in momentum space:

$$\begin{aligned} \langle A_\mu(x) A_\nu(y)\rangle &= \int \frac{d^4p}{(2\pi)^4}\, e^{ip(x-y)}\, \frac{-i\eta_{\mu\nu}}{p^2 - i\epsilon} \\ \langle c(x)\bar{c}(y)\rangle &= \int \frac{d^4p}{(2\pi)^4}\, e^{ip(x-y)}\, \frac{-i}{p^2 - i\epsilon}\,. \end{aligned} \tag{1.3.11}$$

Exercise: Find the exact form of the propagators in a generic R_ξ gauge.

At this advanced stage, one can also add matter fermions to obtain the complete QED action and critically note that the ghost fields can be safely integrated out and completely eliminated from the abelian theory. This is fundamentally because they contribute at most to an overall constant normalization factor that completely disappears when evaluating physically normalized correlation functions. This will decidedly not be the case for non-abelian gauge theories (and also for QED formulated in curved spacetime), but before mathematically addressing the more complex non-abelian case, let us discover and rigorously study a rigid symmetry that is robustly present in the gauge-fixed action and which has far-reaching physical implications, the BRST symmetry.

1.3.2 | BRST Symmetry

Once the gauge symmetry has been fixed, the resulting gauge-fixed action S_{tot} is no longer gauge invariant. This is a necessary consequence, as the gauge-fixing term is explicitly introduced to break this redundancy and define a nonsingular path integral. However, a remnant of the original gauge symmetry survives in the form of BRST (Becchi-Rouet-Stora-Tyutin) symmetry, which is a rigid, global symmetry that depends on a constant, anticommuting (Grassmann-valued) parameter Λ.

Let us consider the total lagrangian in the Feynman gauge:

$$\mathcal{L}_{tot} = -\frac{1}{2}\partial^\mu A^\nu \partial_\mu A_\nu - \partial^\mu \bar{c}\partial_\mu c \tag{1.3.12}$$

Under the BRST symmetry, the fields transform according to the following rules:

$$\begin{cases} \delta_B A_\mu(x) = \Lambda \partial_\mu c(x) \\ \delta_B c(x) = 0 \\ \delta_B \bar{c}(x) = \Lambda \partial^\mu A_\mu(x) \end{cases} \tag{1.3.13}$$

These transformations ensure that the variation of the total lagrangian vanishes up to total derivatives, meaning $\delta_B \mathcal{L}_{tot} = 0$. It is important to recall that the transformation parameter Λ, as well as the ghost fields $c(x)$ and $\bar{c}(x)$, are all Grassmann-valued quantities, which means they mutually anticommute.

A crucial, defining property of this symmetry is that it is nilpotent. Specifically, it is a Lie symmetry featuring a Grassmann odd parameter that satisfies the commutator relation:

$$[\delta_B(\Lambda_1), \delta_B(\Lambda_2)] = 0 \tag{1.3.14}$$

where one evaluates the variation using two different values of the constant Grassmann parameter, Λ_1 and Λ_2.

The nilpotency property becomes much more evident if one defines the Slavnov operator s by formally factorizing out the constant parameter Λ:

$$\delta_B(\Lambda) = \Lambda s\,. \tag{1.3.15}$$

The Slavnov operator s acts as a nilpotent graded variation. In this context, "graded" indicates that the operator anticommutes with Grassmann odd fields, and "nilpotency" explicitly means that it satisfies the condition:

$$s^2 = 0 \tag{1.3.16}$$

Operating twice with s thus gives a vanishing result on any arbitrary field. Explicitly, extracting s from Eq. (1.3.13), we get:

$$\begin{cases} sA_\mu(x) = \partial_\mu c(x) \\ sc(x) = 0 \\ s\bar{c}(x) = \partial^\mu A_\mu(x) \end{cases} \tag{1.3.17}$$

Applying the operator a second time confirms that it indeed gives zero:

$$s^2 A_\mu(x) = s^2 c(x) = 0 \qquad \text{and} \qquad s^2 \bar{c}(x) = \partial^\mu \partial_\mu c(x) = 0 \tag{1.3.18}$$

Notice carefully that the classical ghost equations of motion ($\partial^\mu \partial_\mu c = 0$) have been used in the last equation to ensure the result vanishes. Because one strictly needs to use the equations of motion to satisfy $s^2 = 0$, these particular BRST symmetry rules are said to be nilpotent *on-shell*.

As for any rigid continuous symmetry, Noether's theorem guarantees there is an associated conserved current J^μ_B. This current can be found as usual by applying the Noether trick, which involves artificially extending the global parameter to a local one, $\Lambda \to \Lambda(x)$, in Eq. (1.3.13) and computing the variation of the action:

$$\delta_B S_{tot} = \int d^4x \, (\partial_\mu \Lambda) J^\mu_B \,, \qquad J^\mu_B = -F^{\mu\nu}\partial_\nu c - \partial^\nu A_\nu \partial^\mu c \tag{1.3.19}$$

The associated conserved BRST charge Q_B is simply the spatial integral of the zero-component of this current:

$$Q_B = \int d^3x \, J^0_B = \int d^3x \left(\partial_\nu A^\nu \dot{c} - F^{0i}\partial_i c\right) . \tag{1.3.20}$$

Upon canonical quantization, this classical charge mathematically becomes a nilpotent quantum operator: $Q_B \to \hat{Q}_B$, satisfying $\hat{Q}_B^2 = 0$.

There is an elegant theoretical way of adding auxiliary fields to the lagrangian to make the BRST symmetry nilpotent *off-shell* (i.e., without requiring the equations of motion). To see how this remarkable property emerges, let us consider a Fourier path integral representation[8] of the delta functional utilized during the gauge-fixing procedure:

$$\delta(f(x)) = \int \mathcal{D}B \, e^{-i\int d^4x \, B(x) f(x)} \,, \tag{1.3.21}$$

We then substitute this functional representation directly into the gauge-fixed path integral. Subsequently, we path-integrate over the arbitrary gauge-weighting function h by standard Gaussian integration. The relevant part of the measure manipulations is given by:

$$\begin{aligned} &\int \mathcal{D}h \, \delta\Big(f(A(x)) - h(x)\Big) e^{-i\int d^4x \, \frac{1}{2\xi} h^2(x)} \\ &\qquad = \int \mathcal{D}h \mathcal{D}B \, e^{-i\int d^4x \left(B(x)[f(A(x)) - h(x)] + \frac{1}{2\xi} h^2(x)\right)} \\ &\qquad = \int \mathcal{D}B \, e^{i\int d^4x \left(-B(x) f(A(x)) + \frac{\xi}{2} B^2(x)\right)} . \end{aligned} \tag{1.3.22}$$

From these algebraic manipulations, one naturally finds a new total lagrangian describing the extended set of fields $(A_\mu, c, \bar{c}, B)$. Given that we have previously taken the covariant gauge-fixing function to be $f(A) = \partial^\mu A_\mu$, the lagrangian takes the updated form:

$$\mathcal{L}_{tot} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \partial^\mu \bar{c} \partial_\mu c - B \partial^\mu A_\mu + \frac{\xi}{2} B^2 \,. \tag{1.3.23}$$

[8] This functional representation directly extends the usual one-dimensional formula for the Dirac delta function $\delta(x) = \int \frac{dk}{2\pi} e^{-ikx}$ to the infinite-dimensional functional case.

With the introduction of the auxiliary field, the BRST symmetry transformations now read:

$$\begin{cases} \delta_B A_\mu = \Lambda \partial_\mu c \\ \delta_B c = 0 \\ \delta_B \bar{c} = \Lambda B \\ \delta_B B = 0 \end{cases} \tag{1.3.24}$$

This expanded set of transformations is manifestly nilpotent off-shell. Let us systematically recall the quantum statistics of the fields involved: the fields A_μ and B are commuting (bosonic), while the ghost fields c and $\bar{c}$ are anticommuting (fermionic). The field $c(x)$ is explicitly called the *ghost* and conceptually corresponds to the local gauge parameter $\alpha(x)$, which in a profound sense is turned into a full dynamical field, but endowed with the opposite Grassmann character ($\alpha(x) \to \Lambda c(x)$). The field $\bar{c}$ is analogously called the *antighost*, while the newly introduced field B is the *auxiliary field* (also widely known in literature as the Nakanishi-Lautrup field).

Of course, since B does not contain any derivative kinetic terms, one can easily eliminate it by using its purely algebraic equations of motion:

$$B = \frac{1}{\xi} \partial^\mu A_\mu \tag{1.3.25}$$

Substituting this back into the action, or equivalently, path-integrating over the B field in the functional integral, exactly reproduces the original lagrangian given in Eq. (1.3.12) (up to the ξ dependence). This explicitly checks that the new lagrangian provides a physically equivalent description of the exact same theory. However, eliminating the auxiliary field means that the BRST transformations reduce back to:

$$\begin{cases} \delta_B A_\mu = \Lambda \partial_\mu c \\ \delta_B c = 0 \\ \delta_B \bar{c} = \frac{1}{\xi} \partial^\mu A_\mu \Lambda \end{cases} \tag{1.3.26}$$

where strict nilpotency is unfortunately achieved only on-shell once again.

1.3.3 | Non-Abelian Gauge Fields

Let us now apply what we discussed so far regarding gauge-fixing and quantization to the more complex case of non-abelian gauge fields. Consider the purely bosonic Yang-Mills theory, whose classical dynamics are governed by the lagrangian:

$$\mathcal{L} = -\frac{1}{4} F^a_{\mu\nu} F^{a\mu\nu} \tag{1.3.27}$$

where the non-abelian field strength tensor is explicitly defined as $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu$. This action is invariant under the infinitesimal gauge symmetry transformation given by:

$$\delta A^a_\mu(x) = D_\mu \alpha^a(x) = \partial_\mu \alpha^a(x) + g f^{abc} A^b_\mu(x) \alpha^c(x)\,. \tag{1.3.28}$$

To define a non-singular path integral, one can choose a set of gauge-fixing functions (specifically, one for each generator of the gauge symmetry, indexed by $a = 1, \ldots, \dim G$):

$$f^a(A) = \partial^\mu A^a_\mu \tag{1.3.29}$$

and recognize that, exactly as in the abelian case, the Faddeev-Popov determinant arises directly from its gauge transformation with the local parameter factored out:

$$\frac{\delta f^a(A(x))}{\delta \alpha^b(y)} = \partial^\mu D_\mu^{ab} \delta^4(x-y) \tag{1.3.30}$$

where the covariant derivative acting in the adjoint representation is mathematically written as $D_\mu^{ab} = \partial_\mu \delta^{ab} + g f^{acb} A_\mu^c(x)$. Now, by systematically introducing anticommuting ghost fields to exponentiate this determinant and considering the weighted Lorenz gauge (often referred to as the R_ξ gauge), one finds the complete gauge-fixed action:

$$\mathcal{L}_{tot} = -\frac{1}{4} F^a_{\mu\nu} F^{a\mu\nu} - \partial^\mu \bar{c}^a D_\mu c^a - \frac{1}{2\xi} (\partial^\mu A^a_\mu)^2 \,. \tag{1.3.31}$$

Ghost Interactions and Auxiliary Fields

A crucial distinction arises here: unlike in QED, the ghosts do not decouple from the physical fields. Because a covariant derivative appears in the ghost kinetic term, there is a non-trivial interaction vertex sitting inside the ghost lagrangian:

$$\mathcal{L}_{gh} = -\partial^\mu \bar{c}^a D_\mu c^a = -\partial^\mu \bar{c}^a \partial_\mu c^a - g f^{abc} \partial^\mu \bar{c}^a A^b_\mu c^c \,. \tag{1.3.32}$$

Thus, the ghosts actively enter the Feynman diagrams as internal loop lines, and they will perform their essential theoretical job to guarantee the consistency of the final physical result.

The gauge-fixed lagrangian constructed above possesses a nilpotent BRST symmetry. To study this symmetry off-shell, it is algebraically simpler to introduce the non-dynamical auxiliary fields B^a (the Nakanishi-Lautrup fields) and present the gauge-fixed action in the equivalent form:

$$\mathcal{L}_{tot} = -\frac{1}{4} F^a_{\mu\nu} F^{a\mu\nu} - \partial^\mu \bar{c}^a D_\mu c^a - B^a \partial^\mu A^a_\mu + \frac{\xi}{2} (B^a)^2 \,. \tag{1.3.33}$$

This extended lagrangian is strictly equivalent to the original one. The algebraic equations of motion for the auxiliary field B^a can be solved trivially, yielding:

$$B^a = \frac{1}{\xi} \partial^\mu A^a_\mu \,, \tag{1.3.34}$$

and substituting them back into the lagrangian in Eq. (1.3.33) exactly reproduces Eq. (1.3.31).

Now, incorporating the auxiliary field, the BRST symmetry takes a much simpler and elegant form:

$$\begin{cases} \delta_B A^a_\mu = \Lambda D_\mu c^a \\ \delta_B c^a = -\frac{g}{2} f^{abc} c^b c^c \Lambda \\ \delta_B \bar{c}^a = \Lambda B^a \\ \delta_B B^a = 0 \end{cases} , \tag{1.3.35}$$

which can be rigorously verified to be nilpotent off-shell. To recognize and systematically derive the specific BRST variation of the ghosts c^a, one could have directly imposed the condition of nilpotency of the BRST rule on the gauge field A^a_μ:

$$[\delta(\Lambda_1), \delta(\Lambda_2)] A^a_\mu = 0$$

i.e., expressed in terms of the Slavnov operator s:

$$s^2 A^a_\mu = 0 \,. \tag{1.3.36}$$

Eliminating the auxiliary field B^a by substituting its equation of motion ($B^a = \frac{1}{\xi}\partial^\mu A^a_\mu$) modifies the BRST transformation rule of the antighost to:

$$\delta_B \bar{c}^a = \Lambda \frac{1}{\xi} \partial^\mu A^a_\mu \tag{1.3.37}$$

making the BRST symmetry nilpotent only on-shell (i.e., only when the classical equations of motion are satisfied).

Having successfully found a mathematically consistent gauge-fixed action, one can now use it to generate the full perturbative expansion and the corresponding Feynman diagrams. In particular, one can directly deduce the Feynman rules corresponding to the various propagators and interaction vertices. However, before doing that, it is essential to discuss the underlying BRST quantization method and some of its most profound properties. This framework relies heavily on the fact that the BRST symmetry can be utilized to formally show that physical observables do not depend on the specific gauge chosen. This vital physical property fundamentally justifies why one uses BRST symmetry as a guiding principle in gauge theories.

1.4 | BRST Quantization

BRST quantization is a powerful algebraic method that allows one to systematically find the complete gauge-fixed action entering the path integral, thoroughly incorporating both the necessary ghost fields and the gauge-fixing terms. It is widely recognized by extending the key features observed in simpler examples, and it fundamentally utilizes the continuous BRST symmetry as a robust guiding principle. Eventually, this is a quantization method which is structurally similar to the Faddeev-Popov procedure, but mathematically more rigorous and generalized.

We illustrate the main, essential steps of the method using a non-Abelian Yang-Mills gauge theory as our primary example. The BRST quantization procedure strictly begins with the formal definition of the BRST symmetry. Starting directly from the original gauge-invariant lagrangian, one introduces anticommuting ghost fields physically associated with the original gauge symmetry. The essential, dynamical part of the BRST transformation rules is then determined by strictly imposing the algebraic requirement of nilpotency. Then, to properly implement gauge fixing, one adds an appropriate set of non-minimal auxiliary fields and constructs the complete gauge-fixed total lagrangian by selecting a suitable "gauge-fixing fermion". This elegant framework allows for a wide variety of gauge choices, proving its versatility. The key, indispensable ingredient throughout the entire procedure is the absolute nilpotency of the BRST symmetry operator.

Let us explicitly exemplify this within non-abelian gauge theories. One starts from the standard, classically gauge-invariant lagrangian

$$\mathcal{L} = \frac{1}{2g^2}\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) = \frac{2}{g^2}\mathrm{tr}\left(\frac{1}{4}F_{\mu\nu}F^{\mu\nu}\right) \tag{1.4.1}$$

for the Lie-algebra valued gauge field $A_\mu(x) = -iA^a_\mu(x)T^a$. The classical infinitesimal gauge symmetry of the action is given by the local transformation

$$\delta A_\mu(x) = D_\mu\alpha(x)\,. \tag{1.4.2}$$

At this point, the BRST symmetry on $A_\mu(x)$ is properly defined by introducing the ghost field $c(x) = -ic^a(x)T^a$ and systematically replacing $\alpha(x) \to \Lambda c(x)$ within the infinitesimal gauge transformation, where Λ represents the rigid, anticommuting Lie parameter defining the BRST symmetry[9]. Evidently, the BRST symmetry operating on the gauge field is functionally equivalent to a pure gauge transformation.

The classical infinitesimal gauge transformation and its corresponding BRST transformation are given by:

$$\delta A_\mu(x) = D_\mu\alpha(x) \quad \to \quad \delta_B A_\mu(x) = \Lambda D_\mu c(x) \tag{1.4.3}$$

Proof. The infinitesimal gauge transformation of the gauge field is defined as:

$$\delta A_\mu(x) = D_\mu\alpha(x) = \partial_\mu\alpha(x) + [A_\mu(x), \alpha(x)]$$

To construct the BRST symmetry, we replace the infinitesimal gauge parameter $\alpha(x)$ with $\Lambda c(x)$, where $c(x)$ is the Grassmann-odd ghost field and Λ is a rigid (constant), anticommuting Lie parameter. Substituting this into the covariant derivative yields:

$$\delta_B A_\mu(x) = D_\mu(\Lambda c(x)) = \partial_\mu(\Lambda c(x)) + [A_\mu(x), \Lambda c(x)]$$

[9]The covariant derivative of a field in the adjoint representation is $D_\mu\alpha = \partial_\mu\alpha + [A_\mu, \alpha]$, and similarly $D_\mu c = \partial_\mu c + [A_\mu, c]$. Additionally, the coupling constant is conventionally absorbed into A_μ and c.

Since Λ is a global constant, it factors out of the partial derivative: $\partial_\mu(\Lambda c) = \Lambda\partial_\mu c$. Furthermore, because Λ is a parameter that commutes with the Lie algebra generators T^a (acting only as a scalar coefficient in the color space), we can factor it out of the commutator as well: $[A_\mu, \Lambda c] = \Lambda[A_\mu, c]$. Therefore, we obtain:

$$\delta_B A_\mu(x) = \Lambda\partial_\mu c(x) + \Lambda[A_\mu(x), c(x)] = \Lambda\big(\partial_\mu c(x) + [A_\mu(x), c(x)]\big) = \Lambda D_\mu c(x)$$

which exactly reproduces the right-hand side of Eq. (1.4.3). □

Because it takes the form of a gauge transformation, the classical action is manifestly BRST invariant. Having introduced the dynamical ghost c, we uniquely have to find its specific BRST variation, which is strictly obtained by requiring that the BRST transformation of the physical gauge field A_μ be nilpotent.

The nilpotency of the BRST transformation on the gauge field uniquely determines the BRST transformation of the ghost field:

$$[\delta_B(\Lambda_1), \delta_B(\Lambda_2)]A_\mu(x) = 0 \quad \rightarrow \quad \delta_B c(x) = -c(x)c(x)\Lambda \tag{1.4.4}$$

Proof. It is extremely useful to factor out the constant parameter Λ by introducing the Slavnov operator s, such that $\delta_B = \Lambda s$. Here, s is a Grassmann-odd (anticommuting) derivation, while δ_B is a bosonic (commuting) operator.

Let $\delta_1 \equiv \delta_B(\Lambda_1)$ and $\delta_2 \equiv \delta_B(\Lambda_2)$. Since δ_1 is bosonic, it obeys the standard Leibniz rule. Applying the commutator to A_μ, we find:

$$\begin{aligned}
\delta_1(\delta_2 A_\mu) &= \delta_1(\Lambda_2 s A_\mu) = \Lambda_2\delta_1(sA_\mu) && \text{(since } \Lambda_2 \text{ is constant)} \\
&= \Lambda_2(\Lambda_1 s)(sA_\mu) = \Lambda_2\Lambda_1 s^2 A_\mu = -\Lambda_1\Lambda_2 s^2 A_\mu && \text{(since } \Lambda_1, \Lambda_2 \text{ anticommute)}
\end{aligned}$$

By swapping the indices, $\delta_2\delta_1 A_\mu = \Lambda_1\Lambda_2 s^2 A_\mu$. Thus, the commutator is:

$$[\delta_1, \delta_2]A_\mu = \delta_1\delta_2 A_\mu - \delta_2\delta_1 A_\mu = -2\Lambda_1\Lambda_2 s^2 A_\mu$$

Requiring $[\delta_1, \delta_2]A_\mu = 0$ is therefore strictly equivalent to requiring the nilpotency of the Slavnov operator: $s^2 A_\mu = 0$.

Now, let us compute $s^2 A_\mu$ explicitly. We know that $sA_\mu = D_\mu c = \partial_\mu c + [A_\mu, c]$. Applying s again, and remembering that s acts as an odd derivation (meaning $s(AB) = (sA)B + (-1)^{|A|}A(sB)$), we get:

$$\begin{aligned}
s^2 A_\mu = s(D_\mu c) &= s\big(\partial_\mu c + A_\mu c - cA_\mu\big) \\
&= \partial_\mu(sc) + (sA_\mu)c + A_\mu(sc) - (sc)A_\mu - c(sA_\mu)
\end{aligned}$$

Notice that the terms involving sA_μ form an anticommutator, while the terms involving sc form a commutator:

$$\begin{aligned}
s^2 A_\mu &= \partial_\mu(sc) + [A_\mu, sc] + (sA_\mu)c + c(sA_\mu) \\
&= D_\mu(sc) + \{sA_\mu, c\} = D_\mu(sc) + \{D_\mu c, c\}
\end{aligned}$$

Let us explicitly expand the anticommutator $\{D_\mu c, c\} = (D_\mu c)c + c(D_\mu c)$:

$$\begin{aligned}
(D_\mu c)c &= (\partial_\mu c + A_\mu c - cA_\mu)c = (\partial_\mu c)c + A_\mu c^2 - cA_\mu c \\
c(D_\mu c) &= c(\partial_\mu c + A_\mu c - cA_\mu) = c(\partial_\mu c) + cA_\mu c - c^2 A_\mu
\end{aligned}$$

Summing these two expressions, the $cA_\mu c$ terms cancel out perfectly:

$$\{D_\mu c, c\} = (\partial_\mu c)c + c(\partial_\mu c) + A_\mu c^2 - c^2 A_\mu = \partial_\mu(c^2) + [A_\mu, c^2] = D_\mu(c^2)$$

Substituting this back into our expression for $s^2 A_\mu$, we find:

$$s^2 A_\mu = D_\mu(sc) + D_\mu(c^2) = D_\mu(sc + c^2)$$

For the BRST transformation to be nilpotent ($s^2 A_\mu = 0$) for any arbitrary gauge field configuration, the argument of the covariant derivative must vanish identically:

$$sc + c^2 = 0 \quad \Longrightarrow \quad sc = -c^2$$

Finally, translating this back to the BRST variation $\delta_B = \Lambda s$, we obtain:

$$\delta_B c(x) = \Lambda sc(x) = \Lambda(-c(x)c(x)) = -c(x)c(x)\Lambda$$

which perfectly reproduces the right-hand side of Eq. (1.4.4). □

Note that expanding the matrix product of the ghosts yields:

$$-c(x)c(x) = c^a(x)T^a c^b(x)T^b = c^a(x)c^b(x)T^a T^b = c^a(x)c^b(x)\frac{1}{2}[T^a, T^b] = c^a(x)c^b(x)\frac{i}{2}f^{abc}T^c \tag{1.4.5}$$

In the above sequence, we write the product of generators as a commutator $\frac{1}{2}[T^a, T^b]$ because the Grassmann nature of the ghost fields $c^a(x)c^b(x)$ makes the combination completely antisymmetric in the color indices a, b, so only the antisymmetric part of the generator product survives. This directly implies:

$$\delta_B c^a(x) = -\frac{1}{2}f^{abc}c^b(x)c^c(x)\Lambda\,, \tag{1.4.6}$$

which beautifully reproduces what we intuitively already knew. It explicitly contains the structure constants of the underlying gauge group and precisely coincides with standard geometric derivations. The strict nilpotency of the BRST operator is enough to mathematically prove this last result, which ensures the nilpotency of the BRST symmetry on the ghost is a direct consequence of the Jacobi identity satisfied by the structure constants of the gauge group.

At this stage, the newly constructed BRST symmetry is fundamentally equivalent to the original gauge symmetry. The ghost mathematically does not yet appear in the physical action, and its BRST transformation just formally records the structure constants of the Lie group defining the gauge symmetry. However, we now crucially consider the ghost as a true new dynamical field, meaning the functional path integral extended with the ghost has a chance of being finite:

$$Z = \int \mathcal{D}A\mathcal{D}c\, e^{iS[A]} = \infty \times 0 = \text{finite} \tag{1.4.7}$$

In this formal expression, the zero comes directly from the fact that a Berezin integration over Grassmann variables (the ghost fields c) is identically zero unless the integrand explicitly contains them; here, the classical action $S[A]$ has absolutely no ghost fields in it. Comparing with standard gauge fixing, one may formally identify $\int \mathcal{D}c = \frac{1}{\text{Vol(Gauge)}}$: it vanishes precisely as it represents the mathematical inverse of an infinite gauge volume.

To make the above formula practically useful, one introduces additional real fields, the non-minimal fields $(\bar{c}, B)$, with $\bar{c}$ being fermionic (antighost) and B bosonic (auxiliary field)[10]. These non-minimal fields, endowed with trivial BRST transformations, are specifically introduced to mathematically define the gauge fermion ψ. They carry trivial BRST transformations:

$$\begin{aligned} \delta_B \bar{c}(x) &= \Lambda B(x) \\ \delta_B B(x) &= 0 \end{aligned} \tag{1.4.8}$$

[10] Again with the standard expansion $\bar{c}(x) = -i\bar{c}^a(x)T^a$ and $B(x) = -iB^a(x)T^a$.

which are obviously and manifestly nilpotent. They are used to elegantly define a so-called *gauge fermion* ψ taking the form[11]:

$$\psi = \bar{c} f(A_\mu, B) \tag{1.4.9}$$

where $f(A_\mu, B)$ fundamentally plays the role of the gauge-fixing function. The BRST variation of the gauge fermion (with the constant parameter Λ removed) is manifestly BRST invariant, strictly because the BRST symmetry operator s is nilpotent. The total lagrangian is then defined by:

$$\mathcal{L}_{tot} = \mathcal{L} + s\psi \tag{1.4.10}$$

which leads to a well-defined path integral over all dynamical fields:

$$Z = \int \mathcal{D}A\mathcal{D}c\mathcal{D}\bar{c}\mathcal{D}B\, e^{i\int d^4x\,(\mathcal{L}+s\psi)} \tag{1.4.11}$$

The total effective action is BRST invariant by construction

$$s\mathcal{L}_{tot} = s\mathcal{L} + s^2\psi = 0 \tag{1.4.12}$$

(up to total derivatives, as is usual).

Suitable choices of the function $f(A_\mu, B)$ successfully deliver a gauge-fixed action that possesses a well-defined path integral. It has to be path integrated over the complete set $(A_\mu, c, \bar{c}, B)$ and can then be robustly used to generate the perturbative expansion cleanly in terms of Feynman diagrams. A very useful choice for our specific model is

$$f(A_\mu, B) = \partial^\mu A_\mu - \frac{\xi}{2}B \quad \rightarrow \quad \psi = \bar{c}\left(\partial^\mu A_\mu - \frac{\xi}{2}B\right) \tag{1.4.13}$$

which dynamically implements the (weighted) Lorenz gauge. Recalling the elementary BRST rules conveniently expressed in terms of the Slavnov operator:

$$\begin{cases} sA_\mu = D_\mu c \\ sc = -c^2 \\ s\bar{c} = B \\ sB = 0 \end{cases} \tag{1.4.14}$$

one may directly evaluate the BRST transformation of the chosen gauge fermion and obtain

$$s\psi = B\left(\partial^\mu A_\mu - \frac{\xi}{2}B\right) - \bar{c}\partial^\mu D_\mu c\,. \tag{1.4.15}$$

Then, the correctly gauge-fixed total lagrangian is explicitly:

$$\mathcal{L}_{tot} = \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{c}\partial^\mu D_\mu c + B\partial^\mu A_\mu - \frac{\xi}{2}B^2\,. \tag{1.4.16}$$

Let us thoroughly comment on the full gauge-fixed lagrangian obtained above. It cleanly separates into three distinct conceptual pieces: 1. The classical Yang-Mills kinetic term: $\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$, strictly dictating the physical gauge field dynamics. 2. The gauge-fixing term: $B\partial^\mu A_\mu - \frac{\xi}{2}B^2$. By path-integrating out the non-dynamical auxiliary field B using its algebraic equation of motion $B = \frac{1}{\xi}\partial^\mu A_\mu$, one exactly recovers the standard covariant gauge-fixing term $-\frac{1}{2\xi}(\partial^\mu A_\mu)^2$. 3. The Faddeev-Popov ghost term: $-\bar{c}\partial^\mu D_\mu c$, which intrinsically encapsulates the vital interactions between the unphysical ghosts and the gauge field, precisely compensating for the unphysical longitudinal degrees of freedom propagated by the gauge-fixing term. It explicitly compares successfully with standard results, after properly reinstating the overall normalized trace factor $\frac{2}{g^2}\mathrm{tr}$.

[11] For notational simplicity, we physically neglect an overall $\frac{2}{g^2}\mathrm{tr}(\cdots)$ scaling factor, that must be formally reinstated at the end of the calculation.

Comments

Further important theoretical comments regarding the BRST formalism:

- One can systematically introduce a *ghost number* symmetry by formally assigning ghost number 1 to the ghost c, and ghost number -1 to the antighost $\bar{c}$. All other standard fields inherently have vanishing ghost numbers. The ghost number gracefully helps to classify the various quantities involved: the classical action has ghost number 0, the s operator actively changes the ghost number by 1 (i.e., it carries ghost number 1), and the integrated gauge fermion $\Psi = \int d^4x\, \psi$ structurally has ghost number -1. In particular, external physical states appearing in Feynman diagrams logically must have vanishing ghost number (there is absolutely no need to consider the ghosts as incoming or outgoing asymptotic particles).
- The profound nilpontecy of the BRST symmetry brings in the powerful mathematical concept of *cohomology*. It is rigorously used to correctly identify the physical (i.e., gauge invariant) observables as specific *cohomology classes*: valid physical observables are BRST invariant functions, while any two BRST invariant functions differing merely by the BRST variation of any arbitrary function fundamentally belong to the same class and thus physically identify the exact same physical observable.
- The actions S and $S_{tot} = S + s\Psi$ are physically equivalent and definitively give the same physical results for absolutely any choice of gauge fermion $\Psi = \int d^4x\, \psi$.
- The BRST quantization is a remarkably very general approach. It can be successfully applied to all cases where the underlying gauge algebra possesses constant structure functions and rigorously closes off-shell (meaning no equations of motions are needed to verify the closure of the gauge algebra). For example, it can be seamlessly used to quantize gravity (i.e., the Einstein-Hilbert action), which nevertheless mathematically remains a non-renormalizable theory.
- More general theories, such as local supergravity, are often described by complicated open gauge algebras. Then, even more general mathematical schemes are fundamentally needed to systematically get the correct gauge-fixed action. The powerful Batalin-Vilkovisky method is precisely such a scheme. It structurally embeds the standard BRST method and utilizes further advanced concepts (antifields and antibrackets).

1.4.1 | Cohomology

The profound power of the BRST symmetry stems fundamentally from its nilpotency, a mathematical property that allows us to rigorously define the concept of *cohomology*[12]. The cohomology, or more accurately the cohomology group, is constructed as a vector space whose elements are equivalence classes. In the context of quantum field theory, different equivalent classes are directly identified with different physical observables, precisely corresponding to the truly gauge-invariant observables of the theory. For that purpose, let us make a brief mathematical aside to review the formal concept of cohomology.

Let us consider a general vector space V and a linear mapping operator $\delta : V \longrightarrow V$ such that it squares to zero, $\delta^2 = 0$. Such an operator is formally called *nilpotent*. One defines the kernel of δ,

[12] It should be noted that the equations of motions are often needed to close the algebra of the gauge symmetries, depending on whether the formulation is on-shell or off-shell.

denoted by $\mathrm{Ker}(\delta)$, as the set of all elements $\alpha \in V$ such that they are annihilated by the operator, $\delta\alpha = 0$:

$$\mathrm{Ker}(\delta) = \{\alpha \in V \mid \delta\alpha = 0\}\,. \tag{1.4.17}$$

Its elements are vectors that are mathematically said to be **closed**, and in algebraic topology they are often called "cocycles". Then, one independently defines the image of δ, denoted by $\mathrm{Im}(\delta)$, as the set of all elements $\beta \in V$ such that there exists some predecessor element $\gamma \in V$ for which $\beta = \delta\gamma$:

$$\mathrm{Im}(\delta) = \{\beta \in V \mid \exists\gamma \in V \text{ for which } \beta = \delta\gamma\}\,. \tag{1.4.18}$$

Its elements are vectors that are said to be **exact**, and are often called "coboundaries". Clearly, all exact elements are automatically closed strictly because of the nilpotency property (i.e., applying δ to $\delta\gamma$ yields $\delta^2\gamma = 0$), which logically implies the subspace inclusion $\mathrm{Im}(\delta) \subseteq \mathrm{Ker}(\delta)$. However, the converse is not generally true: not all closed elements may be exact. The cohomology essentially *measures the amount of non-exactness of closed elements*: it is rigorously defined as the set of equivalence classes $[\alpha]$ of closed elements that differ merely by exact elements, i.e.

$$\alpha \sim \alpha' \qquad \text{if} \qquad \alpha' = \alpha + \delta\gamma\,. \tag{1.4.19}$$

The space of these equivalent classes is denoted by the quotient space:

$$H(\delta) = \frac{\mathrm{Ker}(\delta)}{\mathrm{Im}(\delta)} \tag{1.4.20}$$

and is formally called the group of cohomology, or simply cohomology, as visualized in Figure 1.7.

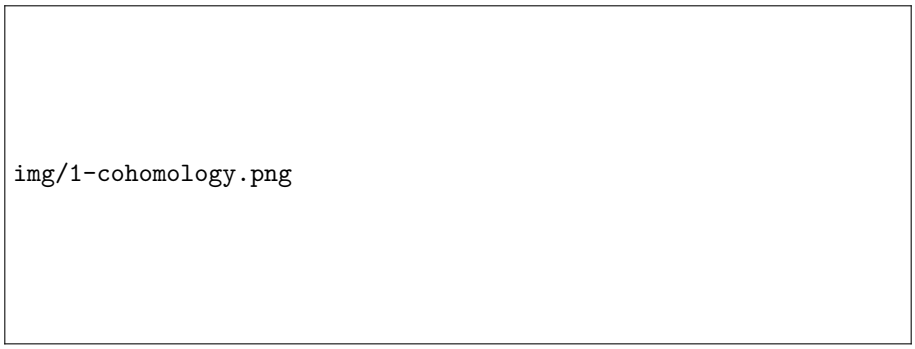


Figure 1.7: The cohomology measures the amount of non-exactness of closed elements. It is mathematically nontrivial when $\mathrm{Im}(\delta) \subset \mathrm{Ker}(\delta)$. The right-hand side specifically depicts the trivial case of vanishing cohomology, $\mathrm{Im}(\delta) = \mathrm{Ker}(\delta)$, where all closed elements are exactly exact.

All the point inside $\mathrm{Ker}(\delta)$ are mapped to zero by the operator δ. The image of δ is a subspace of $\mathrm{Ker}(\delta)$, and the cohomology is the quotient space of these two subspaces: it is the "difference" of $\mathrm{Im}(\delta)$ and $\mathrm{Ker}(\delta)$. When the cohomology is nontrivial, it is mathematically said that there exist closed elements that are not exact. In contrast, when the cohomology is trivial, all closed elements are automatically exact, thus $\mathrm{Im}(\delta) = \mathrm{Ker}(\delta)$ and the quotient space is empty. In this case, the operator δ is said to be *acyclic*.

Application to the BRST Formalism

Returning to the BRST construction discussed previously, the **Slavnov operator** s serves precisely as a nilpotent operator. It acts dynamically on the functionals of the fields (like the lagrangian and

the action), which comprehensively include both ghosts and non-minimal auxiliary fields. *Physical observables* are theoretically defined to be BRST invariant quantities, i.e., they are annihilated by the operator s that defines the infinitesimal BRST transformations (the BRST symmetry perfectly takes the role of gauge invariance within the gauge-fixed theory). Physical observables that differ merely by the BRST variation of some other arbitrary quantity must be identified as physically equivalent: *therefore, physical observables are rigorously defined by the cohomology classes of* s.

For example, the total gauge-fixed action is a nontrivial cohomological element evaluated at vanishing ghost number:

$$S \sim S_{tot} = S + s\Psi\,, \tag{1.4.21}$$

where Ψ represents the gauge-fixing fermion.

Similarly, transitioning to canonical quantization, the classical BRST charge Q becomes a quantum operator $\hat{Q}$ (on the BRST Hilbert space, since it has a redundant structure including ghosts and auxiliary fields). It is the Noether charge fundamentally associated with the BRST symmetry that becomes an operator upon canonical quantization. It intrinsically possesses the fundamental mathematical properties of being: hermitian ($\hat{Q}^\dagger = \hat{Q}$), carrying a ghost number of one, and being strictly nilpotent ($\hat{Q}^2 = 0$). Like all conserved charges, such as the Hamiltonian, it plays a profound dual role:

(i) it is a rigorously conserved quantity due to an underlying Lie symmetry of the system,

(ii) it acts as the algebraic generator of that Lie symmetry on the quantum system under study.

Consequently, *physical states are defined by the cohomology of the operator* $\hat{Q}$ evaluated on the full BRST Hilbert space at vanishing ghost number. That is, **valid physical states** are given by specific vectors of the Hilbert space at vanishing ghost number that are definitively BRST invariant, and thus satisfy the closed condition

$$\hat{Q}\,|\psi_{ph}\rangle = 0.$$

In addition to this, physical states are deemed **equivalent** if they differ solely by the BRST variation of another state: $|\psi_{ph}\rangle$ and $\left|\psi'_{ph}\right\rangle$ are strictly equivalent if they are related by

$$\left|\psi'_{ph}\right\rangle = |\psi_{ph}\rangle + \hat{Q}\,|\chi\rangle\,,$$

for some arbitrary state $|\chi\rangle$. Similarly, *physical operators* are defined by cohomology classes: they are BRST invariant operators, meaning that they commute in a graded sense with the BRST charge $\hat{Q}$, i.e., $[\hat{Q}, \hat{A}_{ph}\} = 0$, and they are subject to an equivalence relation given by

$$\hat{A}'_{ph} \sim \hat{A}_{ph} = \hat{A}_{ph} + [\hat{Q}, \hat{B}\},$$

for some operator $\hat{B}$.

One may mathematically check that the matrix elements of physical operators evaluated between physical states do not depend on the specific representative chosen in the respective classes of equivalence, namely

$$\langle\psi_{ph}|\hat{A}_{ph}|\phi_{ph}\rangle = \langle\psi'_{ph}|\hat{A}'_{ph}|\phi'_{ph}\rangle\,. \tag{1.4.22}$$

To explicitly prove this, for example, if we take $|\phi'_{ph}\rangle = |\phi_{ph}\rangle + \hat{Q}\,|\chi\rangle$, then the modified matrix element expands as

$$\langle\psi_{ph}|\hat{A}_{ph}|\phi'_{ph}\rangle = \langle\psi_{ph}|\hat{A}_{ph}|\phi_{ph}\rangle + \langle\psi_{ph}|\hat{A}_{ph}\hat{Q}|\chi\rangle \tag{1.4.23}$$

but the last term identically vanishes because both $\langle\psi_{ph}|$ and $\hat{A}_{ph}$ are physical (and thus are annihilated by or commute with $\hat{Q}$), so that

$$\langle\psi_{ph}|\hat{A}_{ph}\hat{Q}|\chi\rangle = \langle\psi_{ph}|[\hat{A}_{ph},\hat{Q}]|\chi\rangle + \langle\psi_{ph}|\hat{Q}\hat{A}_{ph}|\chi\rangle = 0 \tag{1.4.24}$$

where we have deliberately taken $\hat{A}_{ph}$ to be bosonic for simplicity, meaning the graded commutator reduces to a standard commutator. Thus, we firmly establish that

$$\langle\psi_{ph}|\hat{A}_{ph}|\phi'_{ph}\rangle = \langle\psi_{ph}|\hat{A}_{ph}|\phi_{ph}\rangle\,. \tag{1.4.25}$$

Finally, it is crucial to note that the full BRST Hilbert space cannot theoretically have a positive-definite norm. For example, exact states constructed like $\hat{Q}\,|\chi\rangle$ inherently have a vanishing norm, as demonstrated by

$$|\hat{Q}|\chi\rangle|^2 = \langle\chi|\hat{Q}^\dagger\hat{Q}|\chi\rangle = \langle\chi|\hat{Q}\hat{Q}|\chi\rangle = 0 \tag{1.4.26}$$

where we have explicitly used the hermiticity of $\hat{Q}$ and its foundational nilpotence. However, this is not a physical problem; it is only theoretically important that the norm restricted to the physical sector of the Hilbert space (the cohomology classes) be positive definite. (*For further details, one may consult chapter 7.4 of Srednicki.*)

1.4.2 | Application: Gauge-Fixing of Perturbative Quantum Gravity

A further compelling application of the BRST method lies in the systematic gauge-fixing of the Einstein-Hilbert action for perturbative quantum gravity. In this fundamental theory, the dynamical field is the metric tensor $g_{\mu\nu}(x)$, which geometrically defines the invariant length element in spacetime:

$$ds^2 = g_{\mu\nu}(x)dx^\mu dx^\nu\,. \tag{1.4.27}$$

The classical dynamics of gravity are governed by the **Einstein-Hilbert action**:

$$S_{EH}[g_{\mu\nu}] = \frac{1}{\kappa^2}\int d^4x\sqrt{g}R(g) \tag{1.4.28}$$

where $g = |\det g_{\mu\nu}|$ denotes the absolute value of the determinant of the metric, $R(g)$ is the Ricci scalar uniquely built from $g_{\mu\nu}$, and the constant $\kappa^2 = 16\pi G$ incorporates Newton's constant G. The parameter κ is naturally taken as the perturbative coupling constant of the theory. Our mathematical conventions follow directly from the standard definition of the covariant derivative ∇_μ, which acts on arbitrary vector fields as:

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\rho}V^\rho\,, \qquad \nabla_\mu W_\nu = \partial_\mu W_\nu - \Gamma^\rho_{\mu\nu}W_\rho$$

where the Christoffel symbols $\Gamma^\rho_{\mu\nu}$, identifying the components of the unique, torsion-free Levi-Civita connection, are expressed purely in terms of the metric:

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})\,.$$

Covariant derivatives inherently do not commute when acting on tensors; this non-commutativity is precisely used to define the fundamental curvature tensors:

$$[\nabla_\mu,\nabla_\nu]V^\rho = R_{\mu\nu}{}^\rho{}_\sigma V^\sigma\,, \quad R_{\mu\nu} = R_{\rho\mu}{}^\rho{}_\nu\,, \quad R = g^{\mu\nu}R_{\mu\nu} \tag{1.4.29}$$

known respectively as the Riemann tensor, the Ricci tensor, and the Ricci scalar. By explicitly calculating the commutator, one finds the explicit formula for the Riemann tensor directly in terms of the connection:

$$\begin{aligned} R_{\mu\nu}{}^{\rho}{}_{\sigma} &= \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma} \\ &= \hat{\nabla}_\mu \Gamma^\rho_{\nu\sigma} - \hat{\nabla}_\nu \Gamma^\rho_{\mu\sigma}, \end{aligned}$$

where in the final line, we wrote a highly useful mnemonic notation in which the symbol $\hat{\nabla}_\mu$ denotes a specialized covariant derivative acting exclusively on the upper index of the associated Christoffel symbols.

If we imagine to send $\Gamma \to \Gamma + \delta\Gamma$, then the corresponding change in the Riemann tensor is given by:

$$\delta R_{\mu\nu}{}^{\rho}{}_{\sigma} = \hat{\nabla}_\mu \delta\Gamma^\rho_{\nu\sigma} - \hat{\nabla}_\nu \delta\Gamma^\rho_{\mu\sigma},$$

where the covariant derivatives $\hat{\nabla}_\mu$ are taken with respect to the original connection Γ. Let us furthermore remark that the Ricci tensor and Ricci scalar are obtained by contracting the Riemann tensor with the metric, which is a nontrivial operation because the metric itself is a dynamical field.

Using all these geometric relations recursively, one successfully finds the Einstein-Hilbert action as a highly non-trivial functional of the metric, whose stationary variation leads directly to Einstein's field equations in vacuum. Crucially, this action is constructed as a true scalar under any arbitrary change of spacetime coordinates, which thus constitutes the fundamental gauge symmetry of the theory.

Diffeomorphism Invariance and Perturbative Expansion

An arbitrary smooth change of coordinates (often referred to formally as a diffeomorphism or reparametrization) mapping old coordinates to new ones:

$$x^\mu \to x'^\mu(x)$$

dynamically induces a corresponding change in the metric tensor components given by the standard tensorial transformation law:

$$g_{\mu\nu}(x) \to g'_{\mu\nu}(x') = g_{\alpha\beta}(x) \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} \,. \tag{1.4.30}$$

Under an infinitesimal change of coordinates continuously parametrized by a small vector field $\xi^\mu(x)$ (this will be the gauge parameter of the theory, with four independent components), the coordinates transform as:

$$x^\mu \to x'^\mu = x^\mu - \xi^\mu(x)$$

the metric transforms at linear order in ξ as:

$$\begin{aligned} \delta g_{\mu\nu}(x) &= g'_{\mu\nu}(x) - g_{\mu\nu}(x) = \xi^\rho(x)\partial_\rho g_{\mu\nu}(x) + \partial_\mu \xi^\rho(x) g_{\rho\nu}(x) + \partial_\nu \xi^\rho(x) g_{\mu\rho}(x) \\ &= \nabla_\mu \xi_\nu(x) + \nabla_\nu \xi_\mu(x) \end{aligned} \tag{1.4.31}$$

as can be directly deduced from expanding the finite transformation rule given above. Consequently, there are exactly four independent gauge functions contained within the infinitesimal vector field ξ^μ.

Perturbatively, to study small fluctuations of gravity around a flat Minkowski background spacetime, one systematically splits the full metric as:

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + \kappa h_{\mu\nu}(x) \tag{1.4.32}$$

where $h_{\mu\nu}(x)$ is the dynamical field whose quantized quanta are physically identified as "gravitons". The complete action, when expanded as a power series in terms of the fluctuation field $h_{\mu\nu}(x)$, contains a leading quadratic part which formally identifies the free graviton propagator (only after an appropriate gauge-fixing procedure), plus an infinite number of interaction vertices:

$$S_{EH}[g_{\mu\nu}] = S_2[h_{\mu\nu}] + \sum_{n=3}^{\infty} S_n[h_{\mu\nu}] \,.$$

The full Einstein-Hilbert action is highly nonlinear in $h_{\mu\nu}(x)$, meaning that interaction vertices of any arbitrary order n are fundamentally present in the theory. These vertices possess a quite complicated algebraic structure, but the free quadratic part is relatively easy to derive (see appendix). However, this quadratic part inherently contains a kinetic differential operator that is strictly non-invertible precisely because of the underlying gauge symmetry shown in Eq. (1.4.31). Thus, one absolutely cannot obtain the propagator for $h_{\mu\nu}$ directly, and a rigorous gauge-fixing procedure is unavoidably required.

BRST Transformations for Gravity

Let us now meticulously apply the BRST method to this system. To start with the BRST quantization of the action in Eq. (1.4.28), we first introduce the fermionic ghost field c^μ by effectively replacing the local gauge parameters $\xi^\mu(x) \to \Lambda c^\mu(x)$, and we logically define the fundamental BRST rule on the full metric $g_{\mu\nu}$ by extending the infinitesimal transformation:

$$\delta_B g_{\mu\nu} = \Lambda(\nabla_\mu c_\nu + \nabla_\nu c_\mu) \,. \tag{1.4.33}$$

The strict mathematical requirement of nilpotency ($s^2 = 0$) uniquely fixes the corresponding BRST transformation rule for the ghost fields themselves:

$$\delta_B c^\mu = \Lambda c^\nu \partial_\nu c^\mu \,. \tag{1.4.34}$$

The direct calculation to prove this is algebraically quite lengthy, but recognizing the underlying geometrical considerations vastly simplifies the task. To that distinct purpose, let us notice that the classical gauge symmetry is mathematically generated by Lie derivatives taken along the vector field ξ^μ (conventionally denoted by $\mathcal{L}_\xi$). This insight gracefully translates into the fact that the transformation in Eq. (1.4.31) may be compactly rewritten as:

$$\delta g_{\mu\nu} = \mathcal{L}_\xi g_{\mu\nu}$$

(this mathematical formulation actually defines the Lie derivative on the metric tensor). More generally, any arbitrary tensor geometrically transforms under the infinitesimal change of coordinates given in Eq. (1.4.31) by its Lie derivative evaluated along ξ^μ. In particular, a simple scalar field ϕ transforms smoothly as:

$$\delta\phi = \mathcal{L}_\xi \phi = \xi^\mu \partial_\mu \phi$$

which would naturally dictate a BRST transformation of the form:

$$\delta_B \phi = \Lambda c^\mu \partial_\mu \phi \,.$$

It is significantly simpler to study the nilpotency condition explicitly on the scalar ϕ in order to uniquely fix the BRST rule of the ghosts. This procedure amounts conceptually to extracting the

fundamental structure constants of the infinite-dimensional algebra of diffeomorphisms. Writing $\delta_B c^\mu = \Lambda s c^\mu$ in terms of the abstract Slavnov variation s, we strictly impose nilpotency:

$$s^2\phi = s(c^\mu\partial_\mu\phi) = sc^\mu\partial_\mu\phi - c^\mu\partial_\mu(c^\nu\partial_\nu\phi) = sc^\mu\partial_\mu\phi - c^\mu(\partial_\mu c^\nu)\partial_\nu\phi$$

that unambiguously fixes the variation of the ghost field to be:

$$sc^\mu = c^\nu\partial_\nu c^\mu \quad \longrightarrow \quad \delta_B c^\mu = \Lambda c^\nu\partial_\nu c^\mu\,. \tag{1.4.35}$$

This specific rule elegantly captures the structure constants carried intrinsically by the commutator of the Lie derivatives:

$$[\mathcal{L}_\xi, \mathcal{L}_\eta] = \mathcal{L}_{[\xi,\eta]} \tag{1.4.36}$$

which precisely forms the underlying gauge algebra of the group of diffeomorphisms. Thus, we have successfully found the complete set of BRST transformations that inherently contain the deep geometrical information on the structure constants of the gauge group:

$$\begin{aligned} \delta_B g_{\mu\nu} &= \Lambda(\nabla_\mu c_\nu + \nabla_\nu c_\mu) \\ \delta_B c^\mu &= \Lambda c^\nu\partial_\nu c^\mu\,. \end{aligned} \tag{1.4.37}$$

Gauge-Fixing and the Total Action

Now, to properly implement a robust gauge-fixing procedure, we add the standard non-minimal fields $\bar{c}_\mu$ (antighost) and B_μ (Nakanishi-Lautrup auxiliary field) which strictly possess trivial BRST transformations:

$$\begin{aligned} \delta_B\bar{c}_\mu &= \Lambda B_\mu \\ \delta_B B_\mu &= 0\,. \end{aligned}$$

A particularly convenient gauge choice utilizes the specific functions:

$$f^\mu = \partial_\nu(\sqrt{g}g^{\nu\mu}) \tag{1.4.38}$$

which, when set to zero, efficiently fix the harmonic gauge (widely known in gravity literature as the de Donder gauge). We thoughtfully use this condition to construct the gauge fermion density:

$$\psi = \frac{1}{\kappa^2}\bar{c}_\mu\left(\partial_\nu(\sqrt{g}g^{\nu\mu}) + \alpha g^{\mu\nu}B_\nu\right) \tag{1.4.39}$$

which mathematically possesses the desirable property of identifying a particularly simple and invertible graviton propagator for a specific tuning of the arbitrary gauge parameter α.

At this final stage, we have successfully constructed the full gauge-fixed action for pure quantum gravity, which is elegantly given by:

$$S_{tot} = S_{EH} + s\Psi \tag{1.4.40}$$

with the integrated gauge-fixing fermion $\Psi = \int d^4x\,\psi$. This total effective action is manifestly BRST invariant by construction. It rigorously serves as the essential starting point for any consistent perturbative treatment of quantum gravity.

1.4.3 | Graviton Propagator in Flat Spacetime

The total gauge-fixed gravitational action derived previously is algebraically quite complex, and standard perturbative calculations in quantum gravity are notoriously difficult.

Let us systematically work out some details by carefully keeping only the quadratic approximation in the metric fluctuation $h_{\mu\nu}$. This approximation directly leads to the perturbative propagator of the graviton (and its corresponding ghosts). Expanding the Einstein-Hilbert action strictly to quadratic order in the fluctuation field $h_{\mu\nu}$, one explicitly finds[13]:

$$S_2[h] = \int d^4x \left[\frac{1}{4}h^{\mu\nu}\partial^2 h_{\mu\nu} - \frac{1}{8}h\partial^2 h + \frac{1}{2}\Big(\partial^\nu h_{\nu\mu} - \frac{1}{2}\partial_\mu h\Big)^2\right], \tag{1.4.41}$$

with the trace defined as $h \equiv \eta^{\mu\nu}h_{\mu\nu}$. Here, the raising and lowering of indices is consistently done using only the flat Minkowski background metric $\eta_{\mu\nu}$. As detailed in the appendix, we have redefined the fluctuation field via $h_{\mu\nu} \to \kappa h_{\mu\nu}$ to absorb the coupling constant, and we use a completely similar redefinition for the corresponding ghosts, auxiliary field, and local gauge parameters to ensure canonical kinetic terms.

This quadratic free action dynamically possesses a gauge symmetry under the specifically linearized version of the full diffeomorphism transformation, which at lowest order cleanly reads:

$$\delta h_{\mu\nu}(x) = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu, \tag{1.4.42}$$

with the lowered parameter defined as $\xi_\mu = \eta_{\mu\nu}\xi^\nu$. The de Donder gauge-fixing function utilized previously smoothly linearizes to the precise condition:

$$f^\mu = -\Big(\partial_\nu h^{\nu\mu} - \frac{1}{2}\partial^\mu h\Big), \tag{1.4.43}$$

which, once successfully inserted into the definition of the gauge fermion

$$\psi = \bar{c}_\mu(f^\mu + \alpha B^\mu)\,, \tag{1.4.44}$$

systematically gives rise to a BRST variation of the form:

$$s\psi = B_\mu(f^\mu + \alpha B^\mu) + \bar{c}_\mu\partial^2 c^\mu \sim -\frac{1}{4\alpha}f^2 + \bar{c}_\mu\partial^2 c^\mu, \tag{1.4.45}$$

where in the very last expression, the algebraic B_μ equations of motion have been explicitly used to eliminate the auxiliary field. By smartly setting the gauge parameter to $\alpha = \frac{1}{2}$, one elegantly obtains the absolute simplest form of the total gauge-fixed action (this specific choice ensures the term containing f^2 will exactly cancel the problematic last term in (1.4.41)), and we cleanly find:

$$S_{2,tot} = S_2[h] + s\Psi = \int d^4x \left(\frac{1}{4}h^{\mu\nu}\partial^2 h_{\mu\nu} - \frac{1}{8}h\partial^2 h + \bar{c}_\mu\partial^2 c^\mu\right). \tag{1.4.46}$$

With this vastly simplified action, it is now a simple, straightforward algebraic matter to derive the momentum-space propagators. The quadratic lagrangian specifically for the graviton field $h_{\mu\nu}$ is elegantly cast as:

$$\mathcal{L}_h = \frac{1}{2}h_{\mu\nu}\tilde{P}^{\mu\nu,\alpha\beta}\partial^2 h_{\alpha\beta}, \tag{1.4.47}$$

in terms of the highly symmetric kinetic tensor:

$$\tilde{P}^{\mu\nu,\alpha\beta} = \frac{1}{4}(\eta^{\mu\alpha}\eta^{\nu\beta} + \eta^{\mu\beta}\eta^{\nu\alpha}) - \frac{1}{4}\eta^{\mu\nu}\eta^{\alpha\beta}, \tag{1.4.48}$$

which, when evaluated in an arbitrary spacetime dimension D, mathematically possesses a well-defined inverse given by:

$$P_{\mu\nu,\alpha\beta} = (\eta_{\mu\alpha}\eta_{\nu\beta} + \eta_{\mu\beta}\eta_{\nu\alpha}) - \frac{2}{D-2}\eta_{\mu\nu}\eta_{\alpha\beta}, \tag{1.4.49}$$

[13] Detailed mathematical steps for this calculation may be found in Appendix B.

that strictly satisfies the defining inversion relation:

$$\tilde{P}^{\mu\nu,\alpha\beta}P_{\alpha\beta,\rho\sigma} = \frac{1}{2}(\delta^{\mu}_{\rho}\delta^{\nu}_{\sigma} + \delta^{\nu}_{\rho}\delta^{\mu}_{\sigma})\,. \tag{1.4.50}$$

The fundamental physical propagators are then given directly by:

$$\begin{aligned} \langle h_{\mu\nu}(x)h_{\alpha\beta}(y)\rangle &= \int \frac{d^Dp}{i(2\pi)^D}\, e^{ip(x-y)}\, \frac{P_{\mu\nu,\alpha\beta}}{p^2 - i\epsilon}, \\ \langle c_{\mu}(x)\bar{c}_{\nu}(y)\rangle &= \int \frac{d^Dp}{i(2\pi)^D}\, e^{ip(x-y)}\, \frac{\eta_{\mu\nu}}{p^2 - i\epsilon}\,. \end{aligned} \tag{1.4.51}$$

These derived formulas are perfectly valid in arbitrary spacetime dimensions D, but of course, for standard physical applications, one can smoothly set directly $D = 4$.

1.5 | QCD and the β-function

[...]

1.5.1 | Feynman Rules for Perturbative QCD

Having successfully obtained the completely gauge-fixed action for non-abelian gauge theories, whether by employing the Faddeev-Popov method or the more general BRST method, one can definitively identify the full QCD gauge-fixed Lagrangian by systematically adding the matter fermions (specifically, the quark fields). Employing an arbitrary covariant R_ξ gauge, one straightforwardly obtains the complete gauge-fixed Lagrangian.

We remind the reader of our chosen conventions for the metric, Clifford algebra, and Lie algebra generators:

$$\eta_{\mu\nu} = \text{diag}(-,+,+,+), \qquad \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \qquad [T^a, T^b] = if^{abc}T^c,$$

together with the explicit definitions of the covariant derivatives acting on the fundamental and adjoint representations:

$$D_\mu \psi^i = \partial_\mu \psi^i - igA^a_\mu (T^a)^i{}_j \psi^j, \qquad D_\mu c^a = \partial_\mu c^a + g f^{abc} A^b_\mu c^c.$$

In what follows, to avoid notational clutter, the specific color index i, strictly associated with the fermion representation, is left entirely implicit.

With these established conventions, the complete gauge-fixed Lagrangian evaluated in an arbitrary R_ξ gauge explicitly reads:

$$\mathcal{L}_{tot} = -\frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu} - B^a\partial^\mu A^a_\mu + \frac{\xi}{2}(B^a)^2 - \partial^\mu \bar{c}^a D_\mu c^a - \sum_{f=1}^{6} \overline{\psi}_f(\gamma^\mu D_\mu + m_f)\psi_f \tag{1.5.1}$$

which can be elegantly rewritten, after systematically eliminating the non-dynamical auxiliary field B^a by substituting its own algebraic equations of motion ($B^a = \frac{1}{\xi}\partial^\mu A^a_\mu$), as:

$$\boxed{\mathcal{L}_{tot} = -\frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu} - \frac{1}{2\xi}(\partial^\mu A^a_\mu)^2 - \partial^\mu \bar{c}^a D_\mu c^a - \sum_{f=1}^{6} \overline{\psi}_f(\gamma^\mu D_\mu + m_f)\psi_f\,.} \tag{1.5.2}$$

This fundamental expression serves as the rigorous starting point for mathematically implementing perturbation theory: the total Lagrangian is systematically split as $\mathcal{L}_{tot} = \mathcal{L}_2 + \mathcal{L}_{int}$, where the strictly quadratic part $\mathcal{L}_2$ unequivocally fixes the free propagators, and the interacting part $\mathcal{L}_{int}$ provides the diverse elementary interaction vertices.

Recalling the explicit expansion of the non-abelian field strength tensor

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu\,, \tag{1.5.3}$$

one explicitly finds, up to total spacetime derivatives that do not affect the action,

$$\mathcal{L}_2 = -\frac{1}{2}(\partial_\mu A^a_\nu)^2 + \left(\frac{1}{2} - \frac{1}{2\xi}\right)(\partial^\mu A^a_\mu)^2 - \partial^\mu \bar{c}^a \partial_\mu c^a - \sum_{f=1}^{6} \overline{\psi}_f(\gamma^\mu \partial_\mu + m_f)\psi_f\,, \tag{1.5.4}$$

and the corresponding interaction terms are grouped as:

$$\mathcal{L}_{int} = -gf^{abc}\partial^\mu A^{\nu a}A^b_\mu A^c_\nu - \frac{g^2}{4}f^{abc}f^{ade}A^b_\mu A^c_\nu A^{\mu d}A^{\nu e} - gf^{abc}\partial^\mu \bar{c}^a A^b_\mu c^c + igA^a_\mu \sum_{f=1}^{6}\overline{\psi}_f\gamma^\mu T^a\psi_f\,. \tag{1.5.5}$$

It is a profound consequence of the underlying gauge symmetry that all interaction vertices are universally controlled by the exact same fundamental coupling constant g.

Propagators

The perturbative momentum-space propagators are mathematically defined as the precise inverses of the "kinetic" terms, i.e., the specific differential operators contained strictly within the quadratic Lagrangian $\mathcal{L}_2$. In a generalized hypercondensed notation (applicable for generic real fields ϕ^i and complex fields $\psi^i, \overline{\psi}_i$, whether they are commuting or anticommuting in nature), one formally has the inversion relations:

$$\begin{aligned} S[\phi] = -\frac{1}{2}\phi^i K_{ij}\phi^j \quad &\rightarrow \quad \langle\phi^i\phi^j\rangle = -iG^{ij} \qquad (K_{ij}G^{jk} = \delta^k_i)\,, \\ S[\psi,\overline{\psi}] = -\overline{\psi}_i K^i{}_j\psi^j \quad &\rightarrow \quad \langle\psi^i\overline{\psi}_j\rangle = -iG^i{}_j \qquad (K^i{}_jG^j{}_k = \delta^i_k)\,. \end{aligned} \tag{1.5.6}$$

Thus, by systematically inverting the kinetic operators extracted from $\mathcal{L}_2$, we rigorously obtain the required momentum-space propagators:

$$\begin{aligned} \langle A^a_\mu(x)A^b_\nu(y)\rangle &= -i\int\frac{d^4p}{(2\pi)^4}\,e^{ip(x-y)}\,\frac{\delta^{ab}}{p^2-i\epsilon}\left(\eta_{\mu\nu} - (1-\xi)\frac{p_\mu p_\nu}{p^2}\right)\,, \\ \langle c^a(x)\bar{c}^b(y)\rangle &= -i\int\frac{d^4p}{(2\pi)^4}\,e^{ip(x-y)}\,\frac{\delta^{ab}}{p^2-i\epsilon}\,, \\ \langle\psi_f(x)\overline{\psi}_f(y)\rangle &= -i\int\frac{d^4p}{(2\pi)^4}\,e^{ip(x-y)}\,\frac{-i\not{p}+m_f}{p^2+m_f^2-i\epsilon}\,, \end{aligned} \tag{1.5.7}$$

with absolutely no sum over the index f (which strictly denotes the distinct quark flavours). In pure momentum space, the corresponding graphical Feynman rules are thus definitively given by:



$$-i\frac{\delta^{ab}}{p^2-i\epsilon}\left(\eta_{\mu\nu} - (1-\xi)\frac{p_\mu p_\nu}{p^2}\right)$$

$$-i\frac{\delta^{ab}}{p^2-i\epsilon}$$

$$-i\frac{-i\not{p}+m_f}{p^2+m_f^2-i\epsilon}$$

Vertices

The interaction vertices evaluated in momentum space are mathematically obtained by systematically Fourier transforming the various interaction monomials contained within $\mathcal{L}_{int}$. Let us therefore briefly revisit our established Fourier transform conventions specifically for the effective action $\Gamma[\varphi] = S_{tot}[\varphi]+\text{loop corrections}$[14], and the intimately related *proper vertices* (which strictly

[14]This object is conceptually identical to the "quantum action" discussed in Srednicki's textbook; see Chapter 21.

correspond to amputated 1PI Feynman graphs) which are mathematically generated by it. When formally replacing the full quantum effective action $\Gamma[\varphi]$ by the tree-level classical action $S_{tot}[\varphi]$, the full proper vertices are naturally replaced by the bare classical vertices directly entering the fundamental Feynman rules.

The effective action $\Gamma[\varphi]$ evaluated for a generic field φ is systematically expanded in a functional Taylor series as:

$$\begin{aligned}\Gamma[\varphi] &= \sum_{n=0}^{\infty} \frac{1}{n!} \int d^D x_1 \dots d^D x_n \, \Gamma^{(n)}(x_1, \dots, x_n) \, \varphi(x_1) \cdots \varphi(x_n) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^D p_1}{(2\pi)^D} \cdots \frac{d^D p_n}{(2\pi)^D} \, \tilde{\Gamma}^{(n)}(p_1, \dots, p_n) \, \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^D p_1}{(2\pi)^D} \cdots \frac{d^D p_n}{(2\pi)^D} \, (2\pi)^D \delta^D(p_1 + \cdots + p_n) \Gamma^{(n)}(p_1, \dots, p_n) \, \tilde{\varphi}(p_1) \cdots \tilde{\varphi}(p_n)\end{aligned} \tag{1.5.8}$$

where the rigorous Fourier transform of the proper vertices with strictly *ingoing momenta* are defined as:

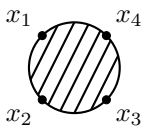
$$\begin{aligned}\tilde{\Gamma}^{(n)}(p_1, \dots, p_n) &= \int d^D x_1 \dots d^D x_n \, e^{i p_1 x_1 + \cdots + i p_n x_n} \, \Gamma^{(n)}(x_1, \dots, x_n) \\ &= (2\pi)^D \delta^D(p_1 + \cdots + p_n) \Gamma^{(n)}(p_1, \dots, p_n)\end{aligned} \tag{1.5.9}$$

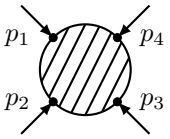
the second line logically following directly from the fundamental translational invariance of the coordinate-space vertex $\Gamma^{(n)}(x_1, \dots, x_n)$ (which guarantees total momentum conservation), while the standard Fourier transform of the fields with explicitly *outgoing momenta* are defined as:

$$\tilde{\varphi}(p) = \int d^D x \, e^{-ipx} \varphi(x) \,, \qquad \varphi(x) = \int \frac{d^D p}{(2\pi)^D} \, e^{ipx} \tilde{\varphi}(p) \tag{1.5.10}$$

From this specific convention, we can clearly see that the spacetime derivative maps as $\partial_\mu \to i p_\mu$ in momentum space, distinctly indicating momentum physically exiting from the field and entering the proper graph $\Gamma^{(n)}(p_1, \dots, p_n)$.

Graphically, these vital correlation functions may be intuitively depicted by:

$\Gamma^{(4)}(x_1, \dots, x_4) =$ 

; $\Gamma^{(4)}(p_1, \dots, p_4) =$ 

(12)

where in the second graph, overall conservation of momentum is implicitly understood, and the external directed lines strictly indicate the physical flow of momentum but crucially do not include the external propagators themselves (i.e., the 1PI graphs are fully amputated).

In particular, analyzing the expansion, one finds that $\Gamma^{(0)}$ is simply the zero-point function (vacuum bubble), $\Gamma^{(1)}(p)$ is the one-point function (tadpole), which is often systematically set to vanish by appropriately adjusting the Vacuum Expectation Value (VEV) of the quantum field $\varphi(p)$, and crucially, $\Gamma^{(2)}(-p, p)$ represents the full effective kinetic term contained in the expansion:

$$\begin{aligned}\Gamma^{(2)}(\varphi) &= \frac{1}{2} \int \frac{d^D p_1}{(2\pi)^D} \frac{d^D p_2}{(2\pi)^D} \, (2\pi)^D \delta^D(p_1 + p_2) \Gamma^{(2)}(p_1, p_2) \tilde{\varphi}(p_1) \tilde{\varphi}(p_2) \\ &= \frac{1}{2} \int \frac{d^D p}{(2\pi)^D} \, \tilde{\varphi}(-p) \Gamma^{(2)}(-p, p) \tilde{\varphi}(p) \\ &= \frac{1}{2} \int \frac{d^D p}{(2\pi)^D} \, \tilde{\varphi}(-p) \left(-p^2 - m^2 + \Pi(p^2) \right) \tilde{\varphi}(p)\end{aligned} \tag{1.5.11}$$

where $-p^2 - m^2$ is exactly the tree-level classical kinetic term whose mathematical inverse (multiplied by i) directly gives the standard perturbative propagator in momentum space $\frac{-i}{p^2+m^2}$, while $\Pi(p^2)$ is identified as the quantum self-energy contribution. The higher-order functions $\Gamma^{(n)}$ for $n \geq 3$ are the true proper interaction vertices.

By systematically replacing the full effective action $\Gamma[\varphi]$ by the tree-level classical action $S[\varphi]$, one cleanly extracts the bare propagator and the bare vertices required for the Feynman rules of the perturbative expansion. In particular, the precise Feynman rules for the bare vertices are extracted directly from iS (since it is e^{iS} that enters the functional path integral) after performing the formal replacement $\Gamma \to S$ in all the above formulas. One robustly obtains the elementary classical vertices $iS^{(n)}$ for a specific vertex involving n fields by systematically considering the formulas for the Fourier transform rigorously derived above: the classical action trivially gives the tree-level component of the full effective action, while the complete effective action contains also the $\hbar$-dependent loop corrections.

A direct plane-wave recipe. A rapid and systematic method exists for obtaining these momentum-space Feynman rules directly from the coordinate-space action. In every interaction monomial, one may simply insert the Fourier expansion for each respective field:

$$\varphi(x) = \int_p e^{ipx} \tilde{\varphi}(p)\,, \qquad \int_p \equiv \int \frac{d^D p}{(2\pi)^D}\,, \qquad \partial_\mu \varphi(x) \to i p_\mu \tilde{\varphi}(p)\,,$$

and subsequently utilize the integral representation of the Dirac delta function to evaluate the spacetime integration over x:

$$\int d^D x\, e^{i(p_1+\cdots+p_n)x} = (2\pi)^D \delta^D(p_1 + \cdots + p_n)\,.$$

Through this procedure, the coefficient of the resulting delta function is identified as the corresponding kernel in the momentum-space action. Multiplying this interaction kernel by an overall factor of i explicitly yields the required Feynman vertex after formally removing the Fourier-transformed fields $\tilde{\varphi}(p)$.

Crucially, if several fields within the interaction monomial are identical (such as the multiple gluon fields in the Yang-Mills action), one must systematically sum over all possible assignments of the labeled external lines to the field slots to correctly account for Bose symmetry. For Grassmann-valued fermionic fields, their strict anticommuting ordering must also be meticulously retained, carefully tracking any relative signs produced by field permutations.

Three-gluon vertex. As a concrete example of this procedure, let us consider the three-gluon interaction term arising from the non-abelian action:

$$S_3 = -g \int d^D x\, f^{abc} (\partial^\mu A^{\nu a}) A^b_\mu A^c_\nu\,.$$

Direct substitution of the three plane-wave expansions immediately gives the momentum-space integral:

$$S_3 = -ig \int_{p_1} \int_{p_2} \int_{p_3} (2\pi)^D \delta^{(D)}(p_1 + p_2 + p_3) f^{abc} p_1^\mu \tilde{A}^{\nu a}(p_1) \tilde{A}^b_\mu(p_2) \tilde{A}^c_\nu(p_3)\,.$$

For instance, explicitly assigning the first, second, and third gauge fields to external lines 1, 2, and 3 respectively—carrying incoming momenta and quantum numbers (p_i, μ_i, a_i)—yields the specific

contribution $-igf^{a_1a_2a_3}p_{1,\mu_2}\eta_{\mu_1\mu_3}$ in S_3, which consequently identifies the term $gf^{a_1a_2a_3}p_{1,\mu_2}\eta_{\mu_1\mu_3}$ within the functional kernel iS_3. Taking the three required functional derivatives with respect to the external fields, or equivalently summing by hand the six possible index assignments due to particle indistinguishability, provides the full symmetrized vertex:

$$\begin{aligned} iS^{a_1a_2a_3}_{\mu_1\mu_2\mu_3} = g\big[&f^{a_1a_2a_3}p_{1,\mu_2}\eta_{\mu_1\mu_3} + f^{a_1a_3a_2}p_{1,\mu_3}\eta_{\mu_1\mu_2} \\ &+ f^{a_2a_1a_3}p_{2,\mu_1}\eta_{\mu_2\mu_3} + f^{a_2a_3a_1}p_{2,\mu_3}\eta_{\mu_2\mu_1} \\ &+ f^{a_3a_1a_2}p_{3,\mu_1}\eta_{\mu_3\mu_2} + f^{a_3a_2a_1}p_{3,\mu_2}\eta_{\mu_3\mu_1}\big]\,. \end{aligned}$$

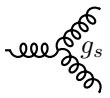
Using the total antisymmetry of the Lie algebra structure constants f^{abc}, this lengthy expression elegantly collapses to:

$$\begin{aligned} iS^{a_1a_2a_3}_{\mu_1\mu_2\mu_3} &= -gf^{a_1a_2a_3}\big[(p_1-p_2)_{\mu_3}\eta_{\mu_1\mu_2} + (p_2-p_3)_{\mu_1}\eta_{\mu_2\mu_3} + (p_3-p_1)_{\mu_2}\eta_{\mu_3\mu_1}\big]\,, \\ &\quad \text{with } p_1+p_2+p_3 = 0\,, \end{aligned}$$

which is precisely the standard three-gluon Feynman rule presented below. This derivative method logically explains why one must not simply keep only the single term in which the derivative acts on line 1: the original interaction monomial contains three identical gluon fields, and functionally differentiating the quantum action with respect to the three external fields intrinsically generates all six unique slot assignments.

Thus, for QCD, the trilinear gluon coupling naturally gives rise to the fundamental interaction vertex $iS^{a_1a_2a_3}_{\mu_1\mu_2\mu_3}(p_1,p_2,p_3)$:

$$\begin{aligned} iS^{a_1a_2a_3}_{\mu_1\mu_2\mu_3}(p_1,p_2,p_3) &= gf^{a_1a_2a_3}p_{1,\mu_2}\eta_{\mu_1\mu_3} + \text{permutations of external lines} \\ &= -gf^{a_1a_2a_3}\Big((p_1-p_2)_{\mu_3}\eta_{\mu_1\mu_2} + (p_2-p_3)_{\mu_1}\eta_{\mu_2\mu_3} + (p_3-p_1)_{\mu_2}\eta_{\mu_3\mu_1}\Big) \\ &= \end{aligned}$$

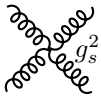


. (1.5.12)

TODO: improve feyn diag

Four-gluon vertex. Similarly, by systematically expanding the quartic gauge field term in the Yang-Mills action and correctly accounting for the $4! = 24$ permutations of the four identical gluon fields, one obtains the 4-gluon vertex rule:

$$\begin{aligned} iS^{a_1a_2a_3a_4}_{\mu_1\mu_2\mu_3\mu_4}(p_1,p_2,p_3,p_4) &= -ig^2 f^{ba_1a_2}f^{ba_3a_4}\eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} + \text{permutations of lines 2 3 4} \\ &= -ig^2\big[f^{ba_1a_2}f^{ba_3a_4}(\eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} - \eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3}) \\ &\qquad + f^{ba_1a_3}f^{ba_4a_2}(\eta_{\mu_1\mu_4}\eta_{\mu_3\mu_2} - \eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4}) \\ &\qquad + f^{ba_1a_4}f^{ba_2a_3}(\eta_{\mu_1\mu_2}\eta_{\mu_4\mu_3} - \eta_{\mu_1\mu_3}\eta_{\mu_4\mu_2})\big] \\ &= \end{aligned} \tag{1.5.13}$$



.

TODO: improve feyn diag

Ghost-antighost-gluon vertex. The interaction term coupling the Faddeev-Popov ghosts to the gauge field within the gauge-fixed lagrangian yields the required ghost-antighost-gluon vertex:

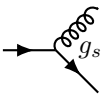
$$iS^{abc}_{\mu}(k,p,q) = gf^{abc}k_\mu\,, \tag{1.5.14}$$

where all momenta stringently follow the ingoing convention utilized above, and k strictly represents the momentum carried by the differentiated antighost field $\bar{c}^a$. It is crucial to note for practical calculations that reversing the direction of this specific momentum physically reverses the displayed sign in the vertex rule.

TODO: add image of ghost-antighost-gluon vertex Feynman diagram

Quark-quark-gluon vertex. Finally, the standard minimal-coupling interaction between the matter fermions and the gauge field provides the fundamental quark-quark-gluon vertex (evaluated for a fixed quark flavour):

$$i(S^{\mu a})^i{}_j(q,p,q') = -g\gamma^\mu (T^a)^i{}_j = \quad . \tag{1.5.15}$$



TODO: improve feyn diag

These non-abelian Feynman rules are considerably more complex than those found in abelian QED, primarily due to the rich self-interaction structure of the gluons mandated by gauge invariance. Nevertheless, armed with these foundational vertices and their corresponding propagators, one may now confidently start computing perturbatively various high-energy scattering processes, systematically renormalize the theory loopwise by introducing appropriate counterterms, and ultimately compute the critical β-function that governs the scale dependence of the strong coupling.

1.5.2 | Perturbative Expansion and the β Function

To rigorously renormalize the theory, one systematically introduces multiplicative counterterms that exactly cancel the ultraviolet infinities arising in loop calculations, effectively matching the renormalization conditions needed to uniquely define the quantum theory in terms of a fixed number of physical observables.

Naively, looking at the separate kinetic and interaction terms of the gauge-fixed QCD lagrangian, one would expect to need at most nine independent renormalization Z factors (we consider here only one flavour of quark for simplicity) as there are exactly 9 independent interaction and kinetic monomials in the fields. We write the fully renormalized lagrangian as follows:

$$\begin{aligned}\mathcal{L} = &-\frac{1}{2}Z_3[(\partial_\mu A^a_\nu)^2 - (\partial^\mu A^a_\mu)^2] - \frac{1}{2\xi}Z_\xi(\partial^\mu A^a_\mu)^2 - Z_{2'}\partial^\mu \bar{c}^a \partial_\mu c^a - Z_2 \overline{\psi}\gamma^\mu \partial_\mu \psi - Z_m m \overline{\psi}\psi \\ &- Z_{3g} g f^{abc}\partial^\mu A^{\nu a} A^b_\mu A^c_\nu - Z_{4g}\frac{g^2}{4} f^{abc} f^{ade} A^b_\mu A^c_\nu A^{\mu d} A^{\nu e} \\ &- Z_{1'} g f^{abc}\partial^\mu \bar{c}^a A^b_\mu c^c + i Z_1 g A^a_\mu \overline{\psi}\gamma^\mu T^a \psi \,.\end{aligned} \tag{1.5.16}$$

which strictly must equal the unrenormalized bare Lagrangian $\mathcal{L} = \mathcal{L}_0$, where all quantities are written in terms of bare fields and bare parameters:

$$\begin{aligned}\mathcal{L}_0 = &-\frac{1}{2}[(\partial_\mu A^a_{0\nu})^2 - (\partial^\mu A^a_{0\mu})^2] - \frac{1}{2\xi_0}(\partial^\mu A^a_{0\mu})^2 - \partial^\mu \bar{c}^a_0 \partial_\mu c^a_0 - \overline{\psi}_0 \gamma^\mu \partial_\mu \psi_0 - m_0 \overline{\psi}_0 \psi_0 \\ &- g_0 f^{abc}\partial^\mu A^{\nu a}_0 A^b_{0\mu} A^c_{0\nu} - \frac{g_0^2}{4} f^{abc} f^{ade} A^b_{0\mu} A^c_{0\nu} A^{\mu d}_0 A^{\nu e}_0 \\ &- g_0' f^{abc}\partial^\mu \bar{c}^a_0 A^b_{0\mu} c^c_0 + i g_0'' A^a_{0\mu} \overline{\psi}_0 \gamma^\mu T^a \psi_0 \,.\end{aligned} \tag{1.5.17}$$

Here we use the standard multiplicative definitions linking the bare fields to the renormalized physical fields:

$$A^a_0 = Z_3^{1/2} A^a \,, \qquad c^a_0 = Z_{2'}^{1/2} c^a \,, \qquad \psi_0 = Z_2^{1/2}\psi \,, \qquad m_0 = Z_m Z_2^{-1} m \tag{1.5.18}$$

and we have provisionally defined four independent bare coupling constants $(g_0, \bar{g}_0, g'_0, g''_0)$. However, gauge invariance (or, more properly and powerfully in this quantized context, BRST invariance) can be rigorously used to derive generalized Ward identities (actually known as Slavnov–Taylor identities in this specific non-abelian context), which conclusively show that the nine Z factors are not all independent: as one might expect from the underlying geometric symmetry, one finds that the bare coupling constants must all be equal, $g_0 = \bar{g}_0 = g'_0 = g''_0$. Also, one can show that there is strictly no need of a divergent counterterm associated to the gauge-fixing term itself, so $Z_\xi = 1$, which algebraically implies that the bare gauge parameter is simply related to the renormalized one by $\xi_0 = Z_3 \xi$.

We denote by μ the $\overline{\text{MS}}$ (Modified Minimal Subtraction) renormalization scale. If μ_{DR} denotes the unrescaled dimensional-regularization mass scale, our specific convention is:

$$\mu^2 = 4\pi e^{-\gamma_E} \mu^2_{\text{DR}} \,. \tag{1.5.19}$$

The Slavnov–Taylor identities then mathematically give the interconnected relations for the bare coupling:

$$g_0^2 = \frac{Z_1^2}{Z_2^2 Z_3} g^2 \mu^\epsilon = \frac{Z_{1'}^2}{Z_{2'}^2 Z_3} g^2 \mu^\epsilon = \frac{Z_{3g}^2}{Z_3^3} g^2 \mu^\epsilon = \frac{Z_{4g}}{Z_3^2} g^2 \mu^\epsilon \tag{1.5.20}$$

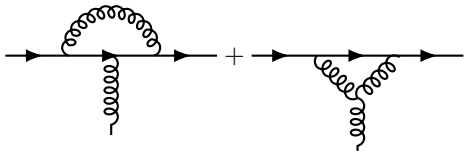
where the spacetime dimension is defined as $d = 4 - \epsilon$, so that the bare coupling g_0 correctly contains the necessary dimensional factor $\mu^{\epsilon/2}$ to keep the action dimensionless. Again, this heavily constrained set of equations implies that there is fundamentally only one independent physical coupling constant governing all interactions. Thus, to compute the crucial β-function, one could simply start by computing just Z_1, Z_2, and Z_3 at the one-loop level and use the first relation in Eq. (1.5.20).

In the Feynman gauge ($\xi = 1$), the one-loop quantum correction to the quark propagator determines Z_2 in the $\overline{\text{MS}}$ scheme[15]:



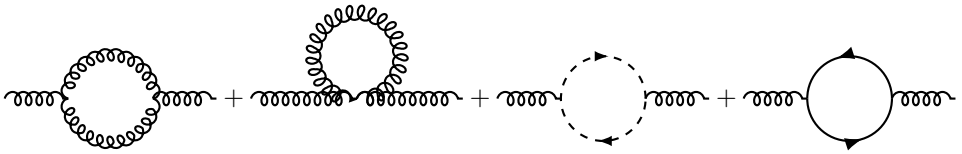
$$\rightarrow \quad Z_2 = 1 - C(R) \frac{g^2}{8\pi^2} \frac{1}{\epsilon} + \mathcal{O}(g^4) \,. \tag{1.5.21}$$

The study of the one-loop radiative corrections to the quark-quark-gluon vertex provides the renormalization constant Z_1:



$$\rightarrow \quad Z_1 = 1 - [C(R) + T(A)] \frac{g^2}{8\pi^2} \frac{1}{\epsilon} + \mathcal{O}(g^4) \,. \tag{1.5.22}$$

Finally, to completely renormalize the gluon wave function and extract Z_3, one must rigorously study all the divergent one-loop contributions to the gluon vacuum polarization propagator:



[15] The group theory factors were described in the appendix of the first set of lecture notes: for $SU(N)$ the index in the fundamental representation is $T(R) = \frac{1}{2}$, while for the adjoint representation it is $T(A) = N$. The quadratic Casimir is universally defined by $T_R^a T_R^a = C(R)\mathbf{1}$ for any arbitrary representation R.

and a laborious, careful calculation encompassing the gluon loops, ghost loop, and fermion loop produces:

$$Z_3 = 1 + \left[\frac{5}{3}T(A) - \frac{4}{3}n_F T(R)\right]\frac{g^2}{8\pi^2}\frac{1}{\epsilon} + \mathcal{O}(g^4)\,. \tag{1.5.23}$$

Here, $T(R)$ and $C(R)$ carefully denote the Dynkin index and the quadratic Casimir invariant, respectively, evaluated specifically in the representation R of the matter fermions, while A strictly stands for the adjoint representation of the gauge bosons.

Now, having evaluated these three fundamental Z factors, one has all the necessary elements to compute the defining one-loop beta function, $\beta(g) = \mu\frac{\partial g}{\partial \mu}$:

$$\beta(g) = -\left[\frac{11}{3}T(A) - \frac{4}{3}n_F T(R)\right]\frac{g^3}{16\pi^2} + \mathcal{O}(g^5)\,, \tag{1.5.24}$$

which, specifically for Quantum Chromodynamics (where $T(A) = 3$ for $SU(3)$ color and $T(R) = \frac{1}{2}$ for the quarks residing in the fundamental representation), dramatically shows the phenomenon of **asymptotic freedom** (i.e., a negative beta function causing the coupling to decrease at high energies) for any number of active quark flavours $n_F \leq 16$.

If interested, these intricate one-loop calculations can be followed in greater technical detail in chapter 73 of Srednicki's textbook.

1.5.3 | Ward Identities

Ward identities arising rigorously from the fundamental BRST symmetry can be systematically used to deduce several crucial properties characterizing the quantum theory. One particular, highly important application is to formally show that physical scattering amplitudes involving external gluons are completely transverse with respect to the specific gluon polarizations. This physical statement mathematically means that if in a given total scattering amplitude M we explicitly extract the physical polarization vector $\epsilon_\mu(p)$ of one of the participating gluons (by setting $M = \epsilon_\mu(p)M^\mu(p)$ for example), then after substituting this specific physical polarization $\epsilon_\mu(p)$ with a purely longitudinal one (i.e. letting $\epsilon_\mu(p) \to p_\mu$) one exactly finds a vanishing result, provided that all the other external states are strictly physical (i.e. they possess on-shell momenta and valid physical polarizations).

Let us briefly sketch how to correctly prove this fundamental result using the underlying BRST symmetry. In the standard LSZ reduction formula, for each asymptotic external gluon state, there is the corresponding gauge field operator $A_\mu(x)$ situated inside the time-ordered correlation function, so that it effectively creates the desired state directly from the vacuum. Moreover, during the reduction procedure, the field $A_\mu(x)$ is systematically processed:

(i) it is acted upon by the corresponding free wave operator $-\partial^2$ (assuming we are operating in the Feynman gauge),

(ii) it is Fourier transformed with the appropriate phase e^{ipx},

(iii) it is contracted with the specific physical polarization vector $\epsilon_\mu(p)$.

Schematically, for an incoming physical state, the creation operator maps as

$$a^\dagger_{in}(p) \to \epsilon^\mu(p)\int d^4x\; e^{ipx}(-\partial^2)A_\mu(x)\,. \tag{1.5.25}$$

Now, deliberately performing the substitution $\epsilon^\mu(p) \to p^\mu$ would algebraically give

$$p^\mu \int d^4x\, e^{ipx}(-\partial^2)A_\mu(x) = -i \int d^4x\, (\partial^\mu e^{ipx})(-\partial^2)A_\mu(x) = i \int d^4x\, e^{ipx}(-\partial^2)\partial^\mu A_\mu(x) \quad (1.5.26)$$

which clearly means that inside the evaluated correlation functions the specific operator $\partial^\mu A_\mu(x)$ explicitly appears. The very last integration-by-parts step is logically most transparent before eliminating the auxiliary field from the action. With the established convention $s\bar{c}^a = B^a$ (up to a possible convention-dependent phase factor of i), the algebraic B-field equation gives directly

$$B^a = \frac{1}{\xi}\partial^\mu A^a_\mu, \qquad \partial^\mu A^a_\mu = \xi\, s\bar{c}^a\,. \quad (1.5.27)$$

Thus, in the Feynman gauge, the divergence operator $\partial^\mu A^a_\mu(x)$ is exactly proportional to the BRST variation of the antighost field $\bar{c}^a(x)$, which in rigorous operator language is correctly generated by acting with the conserved BRST charge $\hat{Q}_B$ on the antighost operator:

$$\partial^\mu A^a_\mu(x) \sim \{Q_B, \bar{c}^a(x)\} \quad (1.5.28)$$

It is therefore physically and mathematically a *cohomologically trivial* operator. Inserting BRST-exact operators into general time-ordered correlation functions may technically produce contact terms, but these specifically do not alter the present physical statement after the full LSZ reduction is applied: the complete on-shell S-matrix element identically vanishes when one external physical gluon polarization is artificially replaced by its corresponding momentum, provided strictly that all the other external asymptotic states remain physical.

Exercises

Sheet 1

[...]

Appendices

A | Notation and Conventions

B | Computations and further developments

Notes on group theory

Lie groups and algebras

Given a simple and compact Lie group G, we conventionally indicate its elements using the exponential parametrization $U(\alpha) = \exp(i\alpha_a T^a)$, where the parameters α_a are continuous and T^a are the infinitesimal hermitian generators that fundamentally satisfy the Lie algebra:

$$[T^a, T^b] = i f^{ab}{}_c T^c \,. \tag{B.1}$$

In general, considering an irreducible unitary representation R of the group G, we automatically obtain a corresponding irreducible representation of its Lie algebra formulated with traceless hermitian matrices T^a_R:

$$[T^a_R, T^b_R] = i f^{ab}{}_c T^c_R \,. \tag{B.2}$$

The representative matrices T^a_R act linearly on a vector space of dimensions $D(R)$, and thus they are explicitly $D(R) \times D(R)$ square matrices. The integer $D(R)$ is technically called the dimension of the representation. Throughout our discussions, we will mostly consider the special unitary group $SU(N)$, whose main fundamental representations are:

- the fundamental (or defining) representation N, which has a dimension $D(N) = N$
- its complex conjugate representation $\overline{N}$, which also has a dimension $D(\overline{N}) = N$
- the adjoint representation Adj, which possesses a dimension $D(\text{Adj}) = N^2 - 1$.

Given a general representation R populated with generators T^a_R, the generators of its complex conjugate representation $\overline{R}$ are mathematically given by

$$T^a_{\overline{R}} = -T^{a\,*}_R \tag{B.3}$$

as can be directly seen from taking the complex conjugate of the original representation's group elements:

$$(\exp(i\alpha_a T^a_R))^* = \exp(-i\alpha_a T^{a\,*}_R) \equiv \exp\Big(i\alpha_a T^a_{\overline{R}}\Big) \,.$$

Killing Metric and Structure Constants

The generators are usually normalized by convention so that in the fundamental representation one explicitly has

$$\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab} \tag{B.4}$$

which structurally fixes the Killing metric, defined generally as $\gamma^{ab} \equiv 2\,\mathrm{tr}(T^a T^b)$, to simply be $\gamma^{ab} = \delta^{ab}$. This metric fundamentally defines scalar products within the algebra and is utilized to seamlessly raise or lower the indices that logically label the generators (which strictly correspond to the indices of the adjoint representation). In particular, it is widely used to formally define the structure constants with purely upper indices:

$$f^{abc} = f^{ab}{}_d \delta^{dc} \tag{B.5}$$

(or more generally expressed as $f^{abc} = f^{ab}{}_d \gamma^{dc}$). This fully raised tensor is proven to be totally antisymmetric. The antisymmetry of f^{abc} is immediately obvious on the first two indices, precisely as seen from the definition of the Lie algebra itself. Then, by carefully using Eq. (B.1) and the normalization Eq. (B.4), one can explicitly compute the following intermediate steps:

$$\begin{aligned}\mathrm{tr}([T^a, T^b]T^c) = i f^{ab}{}_d \,\mathrm{tr}(T^d T^c) = \frac{i}{2} f^{abc} &= \mathrm{tr}(T^a T^b T^c) - \mathrm{tr}(T^b T^a T^c) \\ &= \mathrm{tr}(T^c T^a T^b) - \mathrm{tr}(T^a T^c T^b) = -\mathrm{tr}([T^a, T^c]T^b) = -\frac{i}{2} f^{acb}\end{aligned}$$

so that we find $f^{abc} = -f^{acb}$, which elegantly implies complete antisymmetry across all indices. Note that in the above algebraic manipulations, we have critically used the cyclic property of the trace operation.

The structure constants themselves can be uniquely used to define the adjoint representation 'Adj' by setting the matrix elements as

$$(T^a_{\mathrm{Adj}})^b{}_c = -i f^{ab}{}_c \tag{B.6}$$

since the fundamental Lie algebra relation

$$[T^a_{\mathrm{Adj}}, T^b_{\mathrm{Adj}}] = i f^{ab}{}_c T^c_{\mathrm{Adj}} \tag{B.7}$$

is identically satisfied because of the foundational Jacobi identity.

Representation Indices and Casimir Operators

One mathematically defines the *index* $T(R)$ of an arbitrary representation R by the trace relation

$$\mathrm{tr}(T^a_R T^b_R) = T(R)\delta^{ab}\,. \tag{B.8}$$

The specific index of the fundamental representation N is usually normalized by convention to $T(N) = \frac{1}{2}$, clearly seen in Eq. (B.4). Once this is fixed, the values of the indices corresponding to other representations are determined unambiguously.

Casimir operators are specialized operators built structurally from the generators which commute cleanly with all the generators of the entire group. In particular, the quadratic Casimir operator that is systematically constructed using the Killing metric

$$C_2 = T^a T^b \gamma_{ab} = T^a T^a \tag{B.9}$$

is precisely such an operator. The proof of its commutation is straightforward and simple:

$$\begin{aligned}[C_2, T^b] &= [T^a T^a, T^b] = T^a[T^a, T^b] + [T^a, T^b]T^a = T^a i f^{abc} T^c + i f^{abc} T^c T^a \\ &= i f^{abc}(T^a T^c + T^c T^a) = 0\end{aligned}$$

where the final equality holds directly because the structure constants are completely antisymmetric, while the term in parentheses is symmetric[16]. Since C_2 commutes with all the generators, it fundamentally must be proportional to the identity operator in any given irreducible representation (this is a direct consequence of Schur's lemma). This property mathematically defines the specific number $C(R)$, representing the quadratic Casimir evaluated in the irrep R, by

$$T^a_R T^a_R = C(R)\mathbf{1}\,. \tag{B.10}$$

By cleverly setting $a = b$ in Eq. (B.8) and summing over the indices (i.e., taking the full scalar product utilizing the Killing metric) gives the profound relation

$$C(R)D(R) = T(R)D(\text{Adj})\,. \tag{B.11}$$

For the absolute simplest representations, one analytically finds

$$D(N) = D(\overline{N}) = N \qquad T(N) = T(\overline{N}) = \frac{1}{2} \qquad C(N) = C(\overline{N}) = \frac{N^2-1}{2N} \tag{B.12}$$

$$D(\text{Adj}) = N^2 - 1 \qquad T(\text{Adj}) = N \qquad C(\text{Adj}) = N\,. \tag{B.13}$$

Invariant Tensors and Clebsch-Gordan Coefficients

Finally, it is immensely useful to recall the concept of *invariant tensors*. They are rigorously defined as tensors that remain completely invariant after group transformations are applied. For example, denoting by ψ^i the vectors transforming structurally in the defining representation of $SU(N)$, so that the specific upper index i is dynamically transformed by the defining matrices $U^i{}_j$ of $SU(N)$, then the familiar Kronecker symbol δ^i_j inherently acts as an invariant tensor:

$$\delta^i_j \;\to\; \delta'^i_j = U^i{}_k (U^{-1,T})_j{}^l \delta^k_l = U^i{}_k (U^*)_j{}^l \delta^k_l = U^i{}_k (U^\dagger)_j{}^k = \delta^i_j\,.$$

It geometrically tells us that in combining the representation N with $\overline{N}$ there mathematically must appear a scalar

$$N \otimes \overline{N} = \mathbf{1} \otimes \cdots \tag{B.14}$$

i.e., one can form the invariant scalar $\psi^i \chi_i$ directly out of ψ^i and χ_i. Similarly, the completely antisymmetric tensor carrying N upper indices, $\epsilon^{i_1 i_2 \ldots i_N}$, carefully normalized to $\epsilon^{12\ldots N} = 1$, is a fundamental invariant tensor known also as the Levi-Civita symbol. Indeed, one verifies algebraically that

$$\epsilon^{i_1 i_2 \ldots i_N} \;\to\; \epsilon'^{i_1 i_2 \ldots i_N} = U^{i_1}{}_{j_1} U^{i_2}{}_{j_2} \cdots U^{i_N}{}_{j_N} \epsilon^{j_1 j_2 \ldots j_N} = (\det U)\epsilon^{i_1 i_2 \ldots i_N}$$

but since $\det U = 1$ specifically for $SU(N)$, the invariant property cleanly follows. The exact same logic applies for the lower-index version $\epsilon_{i_1 i_2 \ldots i_N}$.

Other vital invariant tensors are the generators themselves evaluated in any given nontrivial representation R, which we explicitly write as $(T^a_R)^\alpha{}_\beta$, where the upper index α is properly associated with the representation R and the lower index β mathematically corresponds to the conjugate

[16] We have explicitly used the operator identity $[AB, C] = A[B, C] + [A, C]B$ which holds for arbitrary operators.

representation $\overline{R}$[17]. This profound statement follows strictly from the Lie algebra Eq. (B.2) by recognizing that the specific structure constants $f^{ab}{}_{c}$ directly give rise to the generators operating in the adjoint representation that effectively transforms the index a of $(T^a_R)^{\alpha}{}_{\beta}$. This structurally also means that

$$R \otimes \overline{R} \otimes \text{Adj} = \mathbf{1} \oplus \cdots . \tag{B.15}$$

Moreover, since the adjoint is physically a real representation (the $f^{ab}{}_{c}$ are completely real numbers and thus the group elements $e^{i\alpha_a T^a_{\text{Adj}}}$ are strictly real), one dynamically understands that

$$\text{Adj} \otimes \text{Adj} = \mathbf{1} \oplus \cdots \tag{B.16}$$

which perfectly matches with the fact that the established Killing metric δ^{ab} is an invariant tensor critically used to construct scalar products (more generally, the tensor δ^{α}_{β} for the arbitrary representations R and $\overline{R}$ is similarly an invariant tensor). Then combining Eq. (B.15) and Eq. (B.16) logically implies

$$R \otimes \overline{R} = \text{Adj} \oplus \cdots \tag{B.17}$$

which is physically interpreted by saying that $(T^a_R)^{\alpha}{}_{\beta}$ strictly act as Clebsch-Gordan coefficients: they systematically combine the tensors residing in the representation R with those located in the representation $\overline{R}$ in order to cleanly extract a tensor structurally transforming in the adjoint. Said differently, Clebsch-Gordan coefficients are fundamentally invariant tensors.

Finally, let us meticulously define another important invariant tensor, the totally symmetric d^{abc} tensor, cleanly introduced together with the defining anomaly coefficients $A(R)$ by

$$A(R)d^{abc} = \frac{1}{2}\text{tr}\left(T^a_R\{T^b_R, T^c_R\}\right) \tag{B.18}$$

where the overall constant normalization may be conventionally fixed by setting $A = 1$ exclusively for the fundamental representation. It is proven to be totally symmetric and prominently appears in the advanced study of chiral anomalies. The absolute only simple groups that fundamentally have a non-vanishing d^{abc} tensor, and therefore dynamically permit a cubic Casimir operator structurally given by $C_3 \sim d^{abc}T^aT^bT^c$, are $SU(N)$ specifically for $N \geq 3$ and the group $SO(6)$.

Cartan-Weyl basis

It is mathematically often very useful to strategically rewrite the generators characterizing a Lie algebra in the specialized Cartan-Weyl basis. This basis is cleanly defined by first finding the maximal allowable number of generators (or rather, independent linear combinations of said generators) denoted as H_i that strictly commute between themselves:

$$[H_i, H_j] = 0\,. \tag{B.19}$$

This important maximal number is definitively called the *rank* of the group. They are canonically taken to be hermitian operators, and they rigidly define the Cartan subalgebra of the entire Lie algebra. Since they strictly commute, quantum mechanics dictates they can be diagonalized simultaneously in any given matrix representation, and the resulting eigenvalues are mathematically

[17] One may recall from linear algebra that given a representation R, one definitively finds that $R^{-1,T}$, R^*, and $R^{-1,\dagger}$ are also formally representations. These four mathematically distinct representations acts linearly on vectors $v^{\alpha}, v_{\alpha}, v^{\dot{\alpha}}, v_{\dot{\alpha}}$ belonging respectively to the appropriate vector spaces (the original vector space, its rigorous dual, its complex conjugate, and finally the dual complex conjugate). For strict unitary representations, one naturally has $v^{\dot{\alpha}} \sim v_{\alpha}$ and $v_{\dot{\alpha}} \sim v^{\alpha}$.

called the *weights*. This specific definition successfully generalizes the familiar angular momentum generator J_3 characteristic of $SU(2)$, which represents a group possessing a rank of exactly 1. J_3 is precisely the designated generator that is usually diagonalized in standard quantum mechanics[18]. The particular eigenvalues (weights) calculated specifically for the adjoint representation are uniquely called *roots*.

The remaining generic generators are smartly combined into complex combinations so that they logically correspond one-to-one to the roots α_i:

$$[H_i, E_\alpha] = \alpha_i E_\alpha \tag{B.20}$$

which can be cleanly interpreted by physically saying that α_i strictly serve as eigenvalues while the operators E_α structurally act as eigenvectors (the root α acts as a geometric vector possessing components α_i). The specialized generators E_α inherently cannot be hermitian, but rather one can mathematically show that $E_\alpha^\dagger = E_{-\alpha}$, meaning that if α represents a valid root then necessarily $-\alpha$ is logically also a valid root. They perfectly generalize the $J_\pm$ angular momentum step operators traditionally found in $SU(2)$. Finally, one possesses the remaining set of structure constants that logically appear when explicitly calculating:

$$[E_\alpha, E_\beta]\,. \tag{B.21}$$

The underlying Jacobi identity can be systematically used to rigorously study them, and, in particular, one fundamentally finds that

$$[E_\alpha, E_{-\alpha}] = \alpha_i H_i\,. \tag{B.22}$$

which beautifully also generalizes the standard $SU(2)$ case.

This highly structured basis (along with a closely related one technically called the Chevalley basis) proves to be very useful in cleanly deriving the most general properties defining Lie algebras, existing in a close and beautiful analogy with the established theory governing angular momentum in quantum mechanics. In particular, it significantly helps in rigorously constructing the complete mathematical classification enumerating all simple Lie algebras, a towering achievement which is largely due to Killing and Cartan. This elegant classification is visually encoded in the schematic Dynkin diagrams illustrated in fig. 1.

TODO: add image of Dynkin diagrams

The specific algebras explicitly depicted there logically correspond to the following continuous compact groups: $A_n = SU(n+1)$, $B_n = SO(2n+1)$, $C_n = Sp(2n)$, and $D_n = SO(2n)$, where the integer parameter n precisely denotes the rank. The remaining isolated algebras denoted G_2, F_4, E_6, E_7, E_8 physically correspond to the so-called exceptional groups.

[18] Recall the foundational SU(2) algebra: $[J_3, J_\pm] = \pm J_\pm$ and $[J_+, J_-] = 2J_3$.

Perturbative Quantum Gravity

Quadratic Approximation of Einstein-Hilbert Action

To rigorously derive the quadratic approximation of the Einstein-Hilbert action, we must systematically expand the metric tensor around the flat Minkowski background spacetime. The metric is expanded as:

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \tag{B.23}$$

where, for the time being, we have absorbed the perturbative coupling constant κ into the fluctuation field $h_{\mu\nu}$. Then, carefully keeping only terms up to linear order in the fluctuation, one finds the inverse metric and the determinant measures to be:

$$g^{\mu\nu}(x) = \eta^{\mu\nu} - h^{\mu\nu}(x)\,, \quad g = |\det g_{\mu\nu}| = |\det \eta_{\mu\lambda}(\delta^{\lambda}_{\nu} + h^{\lambda}_{\nu})| = 1 + h\,, \quad \sqrt{g} = 1 + \frac{1}{2}h,$$

where the trace is defined as $h \equiv \eta^{\mu\nu}h_{\mu\nu}$, and all spacetime indices are henceforth strictly raised and lowered with the flat background metric $\eta_{\mu\nu}$. The Christoffel symbols linearize directly by substituting (B.23) into their definition:

$$\Gamma^{(1)\rho}_{\mu\nu} = \frac{1}{2}\eta^{\rho\sigma}(\partial_\mu h_{\nu\sigma} + \partial_\nu h_{\mu\sigma} - \partial_\sigma h_{\mu\nu}) = \frac{1}{2}(\partial_\mu h^{\rho}_{\nu} + \partial_\nu h^{\rho}_{\mu} - \partial^\rho h_{\mu\nu})\,, \tag{B.24}$$

and the various geometric curvatures (the Riemann tensor, Ricci tensor, and Ricci scalar, respectively) linearize as follows:

$$\begin{aligned} R^{(1)\rho}_{\mu\nu\ \sigma} &= \partial_\mu\Gamma^{\rho}_{\nu\sigma} - \partial_\nu\Gamma^{\rho}_{\mu\sigma} + \cdots = \frac{1}{2}\partial_\sigma(\partial_\mu h_\nu{}^\rho - \partial_\nu h_\mu{}^\rho) - \frac{1}{2}\partial^\rho(\partial_\mu h_{\nu\sigma} - \partial_\nu h_{\mu\sigma}), \\ R^{(1)}_{\nu\sigma} &= R_{\mu\nu\ \sigma}^{\ \ \mu} = \frac{1}{2}(\partial_\nu\partial^\mu h_{\sigma\mu} + \partial_\sigma\partial^\mu h_{\nu\mu} - \partial_\nu\partial_\sigma h - \partial^2 h_{\nu\sigma}), \\ R^{(1)} &= \partial^\mu\partial^\nu h_{\mu\nu} - \partial^2 h\,. \end{aligned} \tag{B.25}$$

To obtain the proper quadratic approximation of the action, let us first conceptually review how one varies the Einstein-Hilbert action to derive the classical equations of motion. The equations of motion emerge entirely from varying the $\sqrt{g}g^{\mu\nu}$ part within the classical action:

$$S_{EH}[g] = \frac{1}{\kappa^2}\int d^4x\sqrt{g}g^{\mu\nu}R_{\mu\nu}(g), \tag{B.26}$$

producing the first variation:

$$\delta S_{EH}[g] = \frac{1}{\kappa^2}\int d^4x\sqrt{g}\delta g^{\mu\nu}\Big(R_{\mu\nu}(g) - \frac{1}{2}g_{\mu\nu}R(g)\Big)\,.$$

The additional variation coming from the Ricci tensor $R_{\mu\nu}(g)$ in (B.26) produces strictly a total spacetime derivative, as one formally finds:

$$\delta R_{\mu\nu}(g) = \nabla_\rho\delta\Gamma^{\rho}_{\mu\nu} - \nabla_\nu\delta\Gamma^{\rho}_{\mu\rho}. \tag{B.27}$$

This profound relation is widely known as the Palatini identity, which emerges geometrically since the variation of the connection, $\delta\Gamma^{\rho}_{\mu\nu}$, transforms as a true tensor (even though the affine connection $\Gamma^{\rho}_{\mu\nu}$ itself is not a tensor).

To identify the quadratic approximation, we must compute the second variation $\delta^2 S_{EH}[h]$. In this procedure, only the linear variations of the Ricci scalar R and Ricci tensor $R_{\mu\nu}$ are needed, as all other terms involving products of variations will identically vanish when evaluated on the flat

Minkowski background, $g_{\mu\nu}(x) = \eta_{\mu\nu}$. Taking into account the linearized expressions in B.25, and setting the background to $g_{\mu\nu} = \eta_{\mu\nu}$ shortly after, one explicitly finds:

$$\begin{aligned}\delta^2 S_{EH}[h] &= -\frac{1}{\kappa^2}\int d^4x\, h^{\mu\nu}\Big(R^{(1)}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}R^{(1)}\Big) \\ &= \frac{1}{\kappa^2}\int d^4x\, \left(\frac{1}{2}h^{\mu\nu}\partial^2 h_{\mu\nu} - \frac{1}{2}h\partial^2 h - h^{\mu\nu}\partial_\mu\partial^\rho h_{\nu\rho} + h^{\mu\nu}\partial_\mu\partial_\nu h\right) .\end{aligned}$$

Thus, smoothly inserting the symmetry factor $\frac{1}{2}$ arising from the formal Taylor expansion $S[\eta+h] = S[\eta]+\delta S[h]+\frac{1}{2}\delta^2 S[h]+\ldots$, we find that the quadratic order terms sit completely in the second-order functional $S_2 \equiv \frac{1}{2}\delta^2 S_{EH}[h]$:

$$S_2 = \frac{1}{\kappa^2}\int d^4x\, \left(\frac{1}{4}h^{\mu\nu}\partial^2 h_{\mu\nu} - \frac{1}{4}h\partial^2 h - \frac{1}{2}h^{\mu\nu}\partial_\mu\partial^\rho h_{\nu\rho} + \frac{1}{2}h^{\mu\nu}\partial_\mu\partial_\nu h\right) .$$

Finally, systematically redefining the fluctuation field via $h_{\mu\nu} \to \kappa h_{\mu\nu}$ to restore the standard normalization and absorb the gravitational coupling constant, using some further integration by parts to shift derivatives, and appropriately grouping terms into perfect squares, one exactly finds the quadratic action:

$$S_2[h] = \int d^4x\, \left[\frac{1}{4}h^{\mu\nu}\partial^2 h_{\mu\nu} - \frac{1}{8}h\partial^2 h + \frac{1}{2}\Big(\partial^\nu h_{\nu\mu} - \frac{1}{2}\partial_\mu h\Big)^2\right] . \tag{B.28}$$